GFC User Manual

10-143-101 B

Released August 25, 2022

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PRODUCT USAGE STATEMENT

This document is intended to be a general safety overview of Pivotal Systems GFC product and is provided for informational purposes only. Since this sub-system must be properly integrated into the OEM tool, it is the responsibility of the OEM to ensure that all safety regulations and guidelines are properly met.

READ THE MANUAL

Read this entire document and all other publications pertaining to the system, including all relevant manuals, before installing, operating, or maintaining this product.

Practice all plant and product safety instructions and precautions. Failure to follow instructions can cause personal injury and/or property damage. If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired. All personnel who work with or who are exposed to this equipment must take precautions to protect themselves against serious or possibly fatal bodily injury. Pivotal Systems provides information on its products and associated hazards, but it assumes no responsibility for the after-sale operation of the equipment or the safety practices of the owner or user. This equipment produces or uses potentially lethal high voltage and high current. NEVER DEFEAT INTERLOCKS OR GROUNDS.

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1 Introduction

1.1 About This Manual

This manual provides information about the operation and use of the Pivotal Systems standard GFC.

1.2 Technology Background

The GFC product combines two core technologies from Pivotal Systems—a specially designed GFC™ gas flow control valve and a GFM™ gas flow monitor system. The GFC is controlled using one of four standard communication protocols: Analog, DeviceNet, RS-485 or EtherCAT®. This manual deals with Analog, DeviceNet or RS-485communication only. For GFC information relating to the EtherCAT communication protocol, please refer to part number 10-143-104.

1.3 Certifications

The Pivotal Systems GFC is CE marked, and ODVA certified.

1.4 Communications

For any questions on the GFC product and/or the content in this documentation, contact customer support at support@pivotalsys.com or +1-510-770-9125 (selection #4).

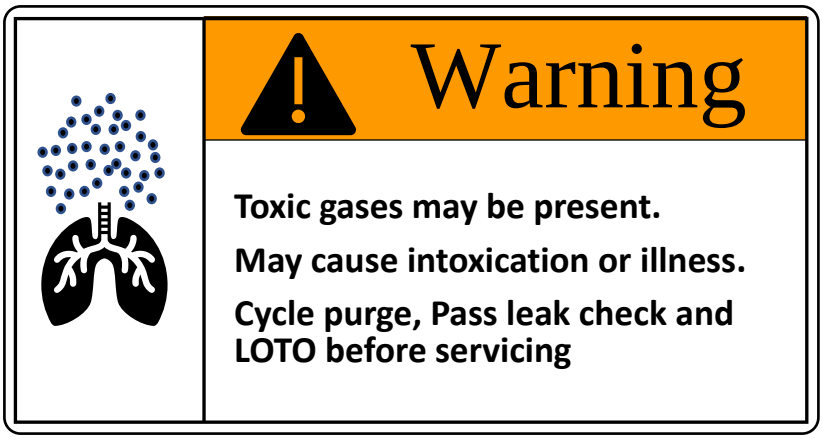
2 General Information

2.1 Hazards and Warnings

WARNING! *Users must have a basic knowledge and understanding of Mass Flow Controllers or Mass Flow Meters operations, in handling hazardous gases, and in performing proper tightening, leak checks, etc.*

WARNING! *Before operating the GFC device, ensure all gas connections have been properly tightened and, where applicable, all electrical connections have been properly terminated.*

WARNING! *Only Pivotal-trained field personnel may service the GFC device. Unauthorized opening of the device will void all warranties and may be unsafe to untrained personnel.*



2.2 Dimensional Information

2.2.1 For 1.125” Gas Stick Substrate

The Figure below is the VCR type of GFC – DeviceNet, Analog & RS485 Communications

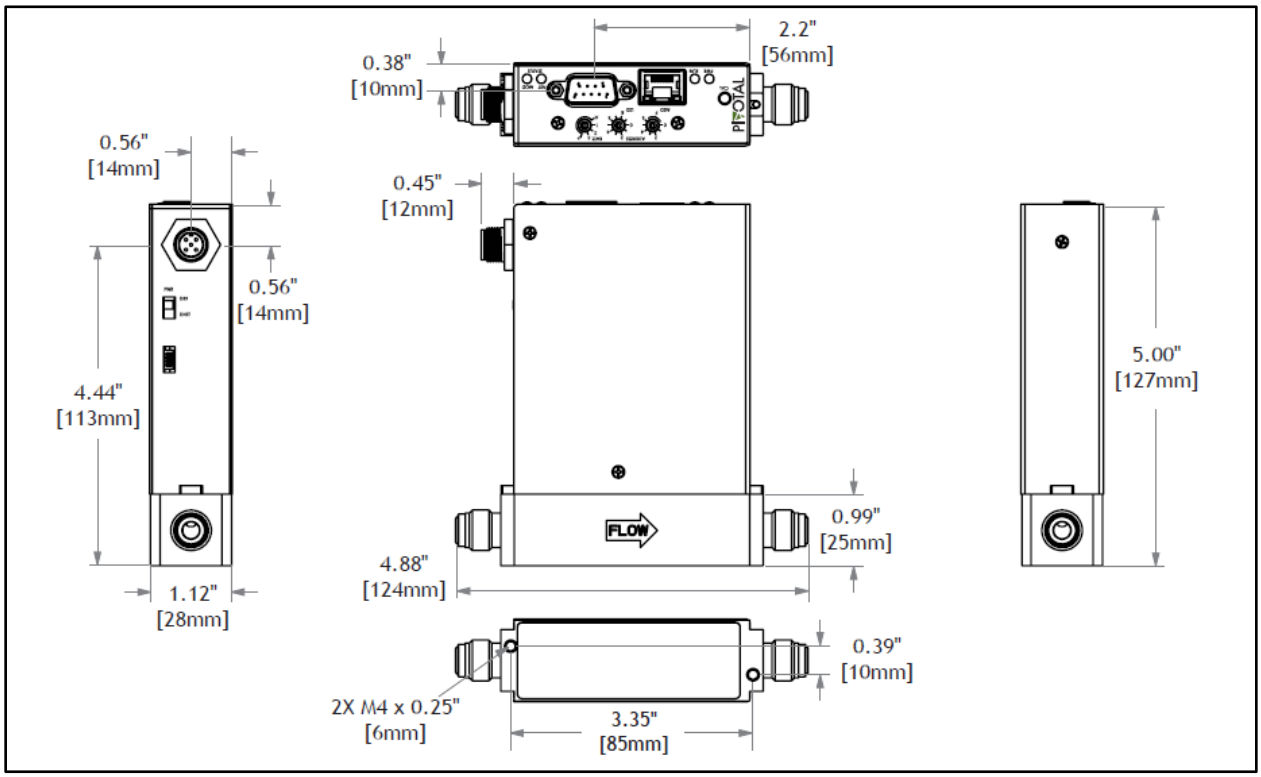


Figure 1: VCR Type of GFC

The figure below is the Surface Mount GFC – DeviceNet, Analog & RS485 Communications.

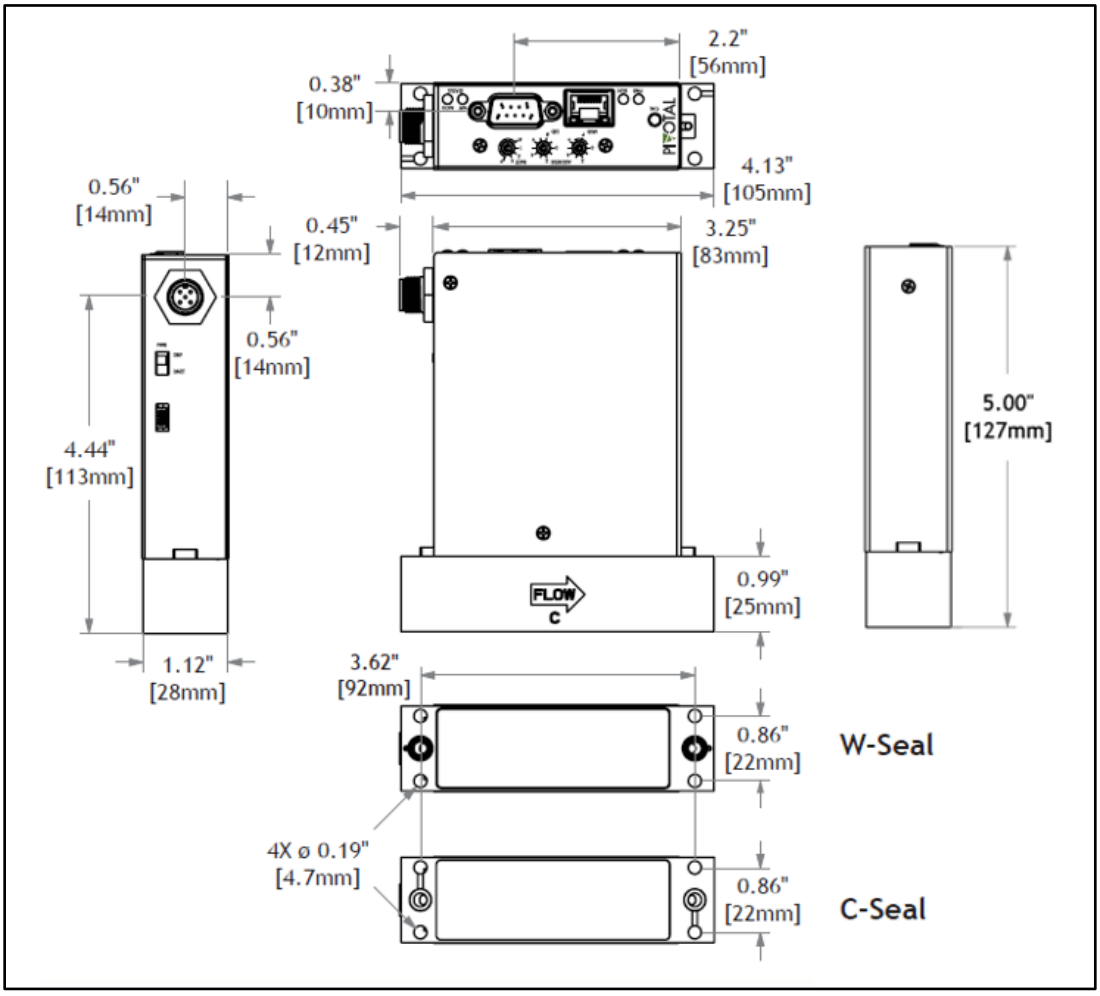


Figure 2: Surface Mount GFC

Depending on the model, the GFC base block is designed for use with the 1-1/8" Surface Mount c-seal or w-seal, or 1.5” VCR gas sticks. For Surface Mount models, spacing and bolt patterns for the GFC are per the SEMI F82-0304 specification for a 1-1/8" mass flow controller. For VCR model, dimensions comply with SEMI F36-0299 Option1.

For 1.5” VCR gas sticks, the dimensions are the same as the 1.125” VCR GFC.

3 Preparation for Installation

3.1 Configuration

The device can be ordered pre-configured for specific applications or in a generic configuration. All devices can be reconfigured for different gases, full scale range, detailed communication settings, etc. Reconfiguring the device can be accomplished using the onboard configuration utility (see section 6.8).

3.2 Gas Stick preparation

Prior to GFC installation, the gas stick from its upstream manual valve to the chamber must be purged of live gas using a nitrogen or argon cycle purge. The purge gas flow should be alternated with a pump-out step so that all live gas is completely pumped out. For inert gases, lines should be pumped and purged for 2–4 hours. For toxic gases may need up to 24 hours of cycle purge. Check with customer facility to determine the exact pump-purge cycles and times. Make sure it passes the leak check and LOTO (lockout/tagout) the gas panel (include all gas sticks) before replacing the GFC.

3.3 Operating Mode

Depending on the model, users may control the GFC via Analog, DeviceNet, or RS-485 protocols. For GFCs that use Analog or RS-485 protocol, slide the selector switch to the “DB9” position. For the DeviceNet protocol, slide the selector switch to the “DNET” position.

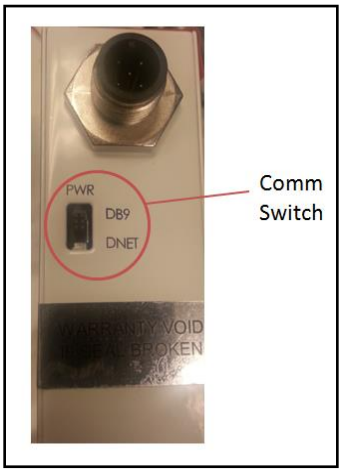


Figure 3: Comm Switch on the Side

3.3.1 Analog Control

Analog controllers may control the GFC using analog protocols as described in Section **8.1**. To use the analog control mode, set the COMM selector to the **DB9** position (refer to Figure 4) and set the Data Rate switch on top of the GFC to **A** (**4**). Analog control is available via the 9-pin D-sub connector at the top of the GFC.

In the analog control mode, Node Address switches are not used.

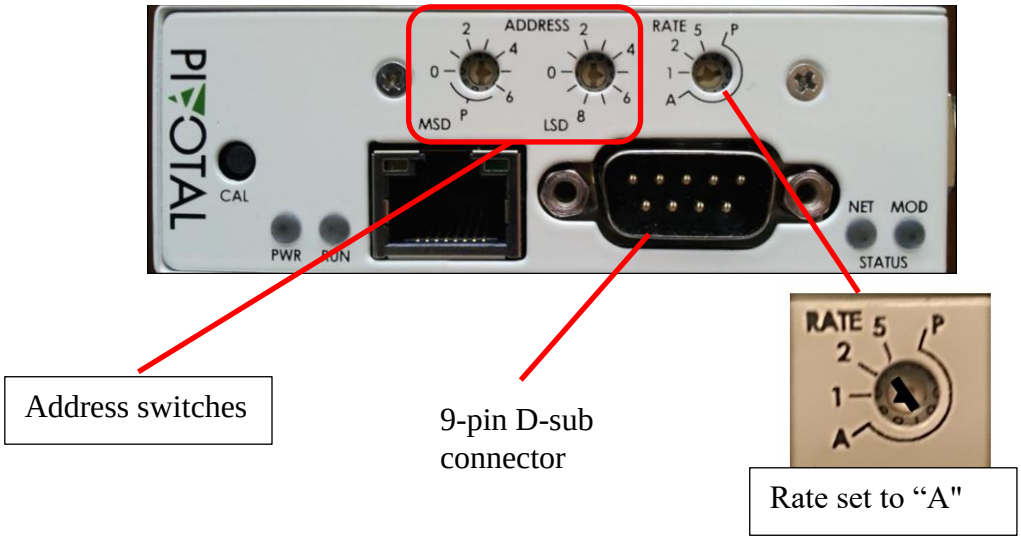


Figure 5: Switches and Address Setup

CAUTION:

Address and Rate switches employ detents to align the switch with the appropriate contact. Switches must be set so that they are in the detent. If they are set between detents, they may provide erratic or incorrect values. The switch arrow may not line up perfectly with silkscreen on covers when the detent is met.

For some tools, a DB9 adapter cable may be required to allow good connection to the GFC to prevent clearance issues with the gas panel door or to ensure correct grounding.

3.3.2 DeviceNet Control

The 1.125” GFC supports the ODVA specification. See section 8.2 DeviceNet Connection. To configure the GFC for DeviceNet communication, set the COMM selector switch to the **DNET** position.

3.3.2.1 DeviceNet Switches: Node Address (MAC ID) Switches

GFC devices have preset switch settings. For DeviceNet, they ship with the MSD Node Address switch set to position **6** and the LSD Node Address switch set to position **3**. These settings correspond to a MAC ID of 63, which is the default for all devices as defined by the ODVA *DeviceNet Specification*. Configure these switches to the appropriate MAC ID for device or to **P** for the tool host to set the MAC ID.

3.3.2.2 DeviceNet Switches: Data Rate Switch

Use the Data Rate switch to configure the baud rate of the device according to the following list:

- Position 1: 125,000 baud
- Position 2: 250,000 baud
- Position 5: 500,000 baud
- Position P: Tool (host) set baud rate

The default baud rate on the GFC is 500,000 baud.



CAUTION:

Address and Rate switches employ detents to align the switch with the appropriate contact. Switches must be set so that they are in the detent. If they are set between detents, they may provide erratic or incorrect values. The switch arrow may not line up perfectly with silkscreen on covers when the detent is met.

3.3.3 RS-485 Control

The GFC may be controlled using the RS-485 communication protocol as described in section **8.3.** To use the GFC in the RS-485 mode, set the COMM selector to the **DB9** position.

3.3.3.1 Node Address (MAC ID) Switches

The MSD Node Address switch corresponds to the ‘10’s value for the MAC ID, and the LSD switch corresponds to the ‘1’s value for the MAC ID.

GFC devices have preset switch settings. For RS-485, GFC’s ship with the MSD Node Address switch set to position **3** and the LSD Node Address switch set to position **3**. These settings correspond to a MAC ID of 33. Configure these switches to the appropriate MAC ID for the device. A position of MSD to **0** and LSD to **0** allows the tool (host) to set the MAC ID and baud rate.

For non-FNET tools where the manufacturer requests MAC IDs starting from 1, contact Pivotal Field Service to adjust the settings.

3.3.3.2 Data Rate Switch

Use the Data Rate switch for configuring the baud rate of the device. The Data Rate switch sets the baud rate according to the following list:

- Position 1: 9600 baud
- Position 2: 38,400 baud
- Position 5: 115,200 baud

The default baud rate on the GFC for RS-485 is 115,200 baud.



CAUTION:

Address and Rate switches employ detents to align the switch with the appropriate contact. Switches must be set so that they are in the detent. If they are set between detents, they may provide erratic or incorrect values. The switch arrow may not line up perfectly with silkscreen on covers when the detent is met.

4 User Interface Setup Preparation

4.1 IP Address Setup

The GFC has a GUI to allow users to view and change configuration parameters on the GFC. The laptop or computer only needs to set up once for all Pivotal GFCs.

4.1.1 Set the computer IP address to 192.168.11.10 and subnet mask to 255.255.0.0.

4.1.1.1 If using Windows 10, go to Settings and select “Network and Sharing Center” and then select “Ethernet”.

4.1.1.2 Select “Change adapter settings”.

4.1.1.3 Right click on the Ethernet adapter being used, select “Properties”.

4.1.1.4 Highlight “Internet Protocol Version 4 (TCP/IPv4)” and click the Properties button.

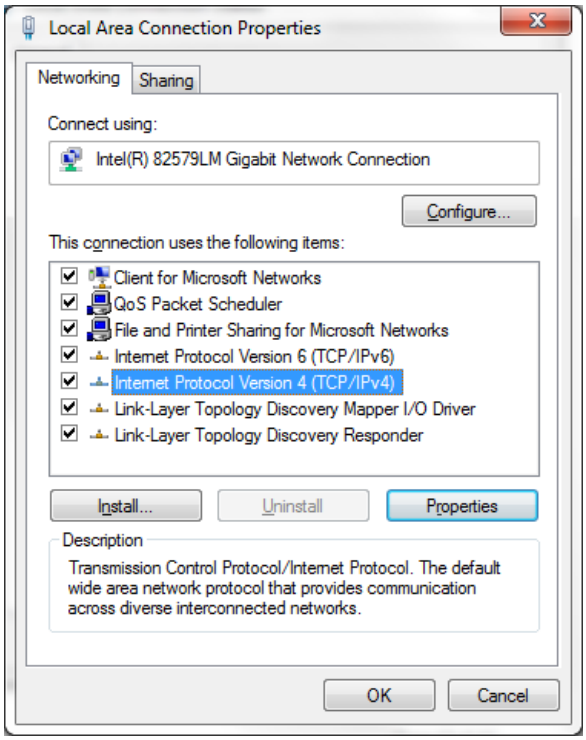


Figure 6: IP Configuration

4.1.1.5 Set the IP address to 192.168.11.10 and subnet mask to 255.255.0.0.

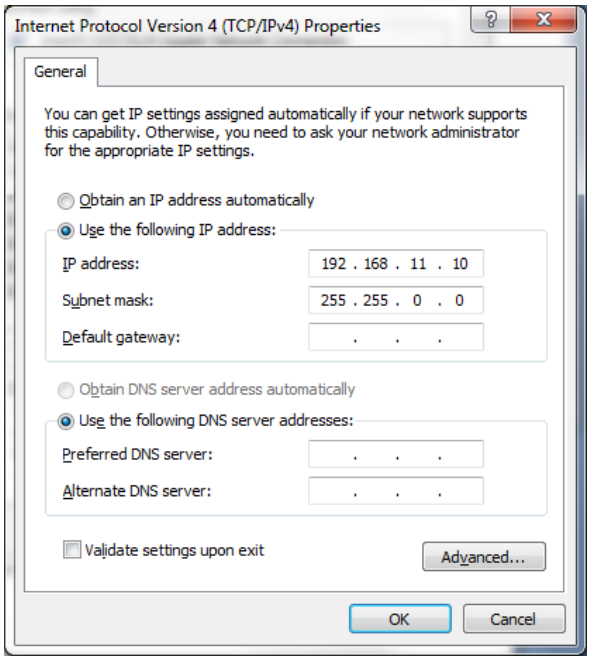


Figure 7: IP and Subnet Mask

5 Installing GFC on Gas Stick

5.1 W-Seal and C-Seal Interfaces

To install the GFC with a w-seal or c-seal interface:

5.1.1 Remove the protective tape from the GFC upstream and downstream gas ports.

Note:

Do not place the GFC seal side down on any surface once the tape has been removed. Doing so has the potential to introduce particles to the GFC and to the gas stick.

5.1.2 Ensure that seal material is appropriate for the gas. Incorrect material selection can result in corrosion, contamination, and leaks.

5.1.3 Place appropriate seal on the GFC at the upstream and downstream gas ports. Install the seals thru the anti-galling coated bolts screws to hold them in place before installing to the gas stick substrate.

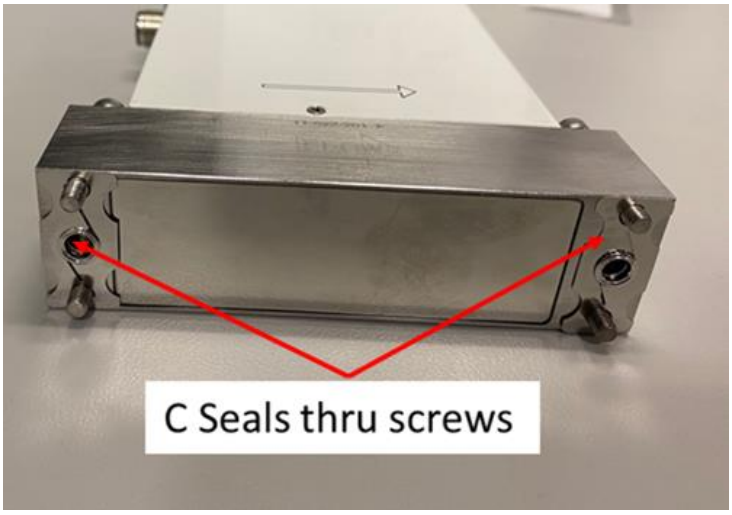


Figure 8: Seals Installation

5.1.4 Tighten the 4 screws evenly (2 turns a time) by an x-pattern. For 1.125” substrates, torque these bolts to 3.5–3.8 N-m (31 – 34 in-lbs).

5.2 VCR Interface

To install the GFC with a VCR interface:

5.2.1 **Place VCR gaskets between VCR glands.**

5.2.2 **Ensure that seal material is appropriate for the gas. Incorrect material selection can result in corrosion, contamination, and leaks.**

5.2.3 **Use wrenches to tighten these VCR nuts.**

5.2.3.1 For nickel and stainless steel gaskets, apply 1/8 turn past finger tight. For copper gaskets, apply 1/4 turn past finger tight.

6 Powering On the GFC

6.1 Electrical Connection

To install the GFC to the electrical connection:

6.1.1 For Analog communication, plug-in the DB9 connector at the top of the GFC and make sure the Data Rate switch is set to A.

6.1.2 For DeviceNet, plug-in the DeviceNet cable to the side of the GFC.

6.1.3 For RS-485 communication, plug-in the DB9 connector at the top of the GFC. Make sure the MSD address and LSD address, as well as Data Rate switches, are set appropriately.

After connecting the cables, the GFC will power on. The power LED will glow solid green . If this light stays off, there is a power delivery problem. The run light will flash green when the GFC has successfully completed startup.

6.2 Leak Check

Check sealing surfaces for leaks using either rate of rise or helium leak check. The leak rate should be less than 10^{-9} mbar-L/sec, (Some may choose mtorr/minute based on the chamber volume). The method chosen should be dictated by the facility.

Warning: No gas can be turned on (flow to the GFC) until the whole gas delivery line passed leak check.

6.3 Pump-Purge

The Pump-Purge process allows any ambient moisture introduced during installation to be eliminated.

Use pump and purge as per the OEM or facility procedure using nitrogen, argon, or other process purity inert gas. The stick should be pumped and purged for a minimum of 2 hours for an inert gas and a minimum of 4 hours for a corrosive gas. Refer to the facility or tool procedures for specific requirements. One must do a leak check (External) to make sure the new installation is leak tight.

6.4 Warm-Up: 30 Minutes

The GFC requires a 30-minute warm-up time. When operated in a room temperature environment, the internal temperature of the GFC is approximately 40°C.

6.5 Heating

The normal internal operating temperature of the GFC, as indicated by the internal temperature sensor feedback, is approximately 40°C. Use external heaters only if your application requires a higher temperature. Heaters should only be applied to the base of the GFC; the GFC case should be free of heaters. Monitor the temperature feedback from the GFC to ensure it does not exceed the maximum temperature of 60°C.

6.6 Operating Pressure

The GFC is designed for supply pressures typically from 15–45 psig (some gases can go lower) and for downstream pressures between vacuum (mtorr range) and 400 Torr (absolute). As long as supply and downstream pressures are within these stated guidelines, there is no additional restriction regarding the pressure drop across the GFC.

6.7 Connecting to the GFC Configuration Utility

The GFC has a GUI to allow users to view and change configuration parameters on the GFC.

6.7.1 Connect an Ethernet cable to the GFC Ethernet port and computer.

6.7.2 Open a web browser

Chrome, Microsoft Edge and Firefox all work with the GFC software. Any of these browsers can be downloaded for free. They are available at:

- Chrome: www.google.com/chrome/
- Microsoft Edge: https://www.microsoft.com/en-us/edge
- Firefox: www.mozilla.com

6.7.3 Enter the GFC IP address (192.168.12.38) into the address bar of the browser.

The GFC Status page will display current GFC settings.

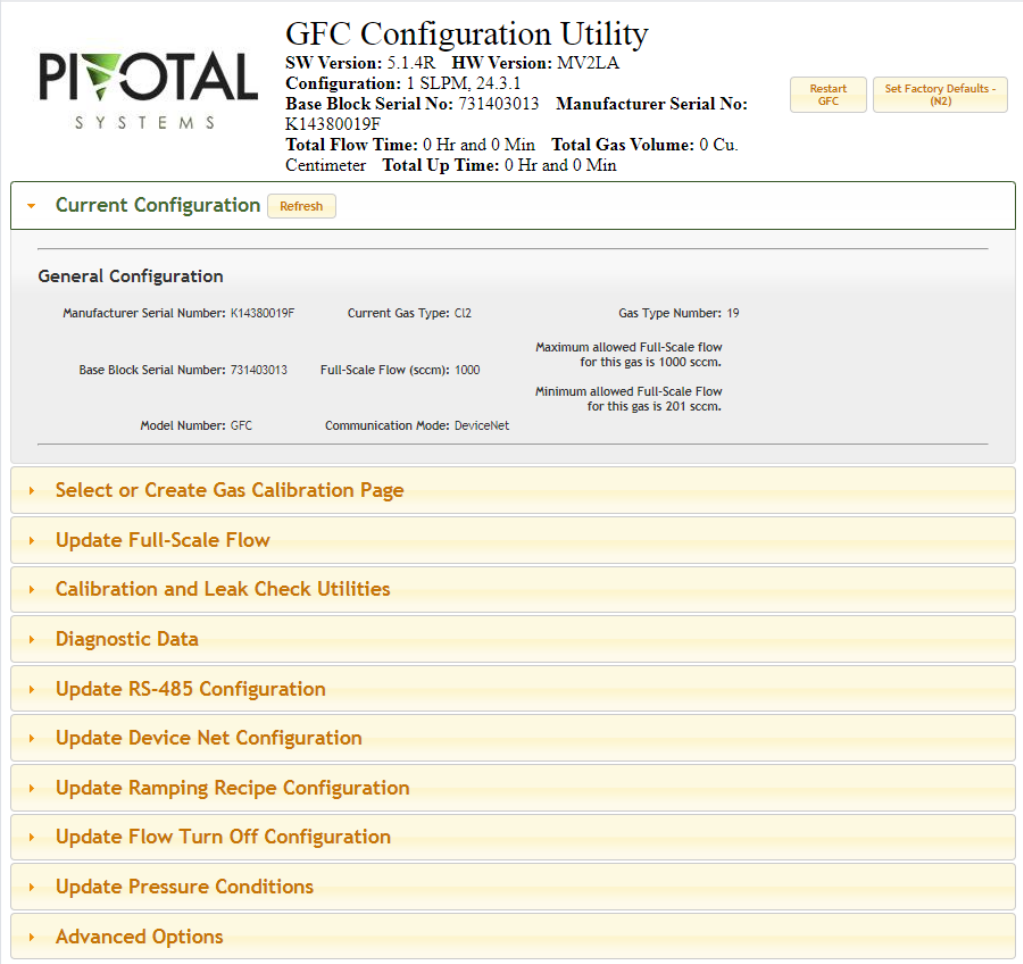


Figure 9: User Interface Screen

6.8 GFC Configuration Utility

The configuration GUI contains information about the device in the header as well as several tabs which allow various changes to the device settings.

6.8.1 Header Information

The header contains a variety of information:

1. SW Version – Displays current firmware version

2. HW Version – Displays internal hardware version information
3. Baseblock Serial Number – Displays the permanently marked serialized value
4. Manufacturer Serial Number – Displays the manufacturer configured serial number
5. Total Flow Time – Displays a running odometer of flow time
6. Total Gas Volume – Displays an approximate quantity of gas flow through the device
7. Restart – Button to restart the GFC (wait 30 seconds to reestablish communication)
8. Set Factory Defaults – Resets the configuration of the device to the default configuration of N2 gas and the maximum Full Scale flow for that bin.

6.8.2 Current Configuration

The Current Configuration section of the GFC Configuration Utility provides more detail about the GFC's settings.

6.8.2.1 Refresh Button

Clicking on this button will update the information in the Current Configuration section.

6.8.2.2 General Configuration

This section provides information as to the currently configured gas, SEMI gas code, currently configured full scale value, maximum and minimum allowed full scale value for the configured gas on this device, and the communication mode.

6.8.3 Selecting Gas Type and configuring flow range

By default, GFC unit can be configured with up to 254 unique gas pages in its gas list. Some tools can only handle devices with 2 to 8 gas pages.

"Select or Create Gas Calibration Pages" section in conjunction with "Update Full Scale Flow" section allows customer to

6.8.3.1 Select a gas from existing Gas list to match the process gas and required maximum flow.

6.8.3.2 Create a limited list (Custom Gas list) of gas pages and specify the Active Gas Instance

6.8.3.3 Update the existing Custom Gas list

6.8.3.4 Modify Full Scale Flow (Flow Range) of existing Gas

6.8.4 How to identify if you have Custom Gas list or complete gas list

6.8.4.1 Select the dropdown menu for “Select or Create Gas Calibration Page” to access the fields that allow change to the gas configuration.

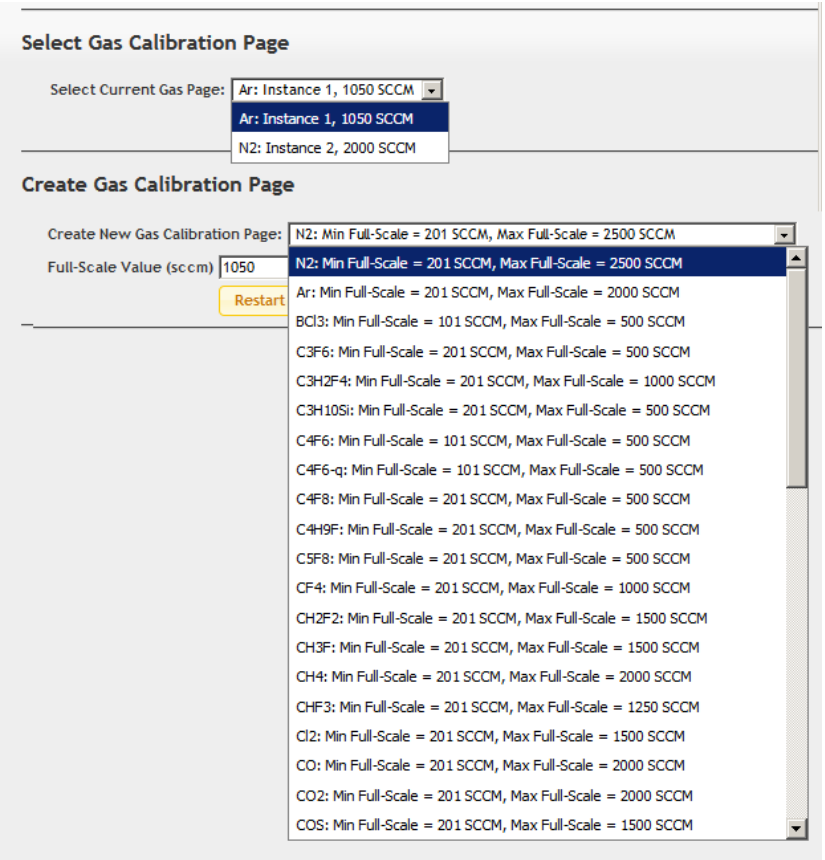


Figure 10: Typical Gases shown in each field

6.8.4.2 Check “Create Gas Calibration Page” section, “Create New Gas Calibration Page” drop down list. It contains the list of gases available on the selected GFC Unit

6.8.4.3 Check “Select Gas Calibration Page” section. If the list of gases available from “Select Current Gas Page” drop down is 2-3 items long or is shorter than the list of gases in the “Create New Gas Calibration Page” list, the unit is configured for Custom Gases.

6.8.5 How to select the Active gas (Current Gas Type) from the existing gas list

6.8.5.1 Select the dropdown menu for “Select Current Gas Calibration Page” to select the desired gas and flow size.

6.8.5.2 Click “Restart and Apply Change” button to activate the selected Gas instance.

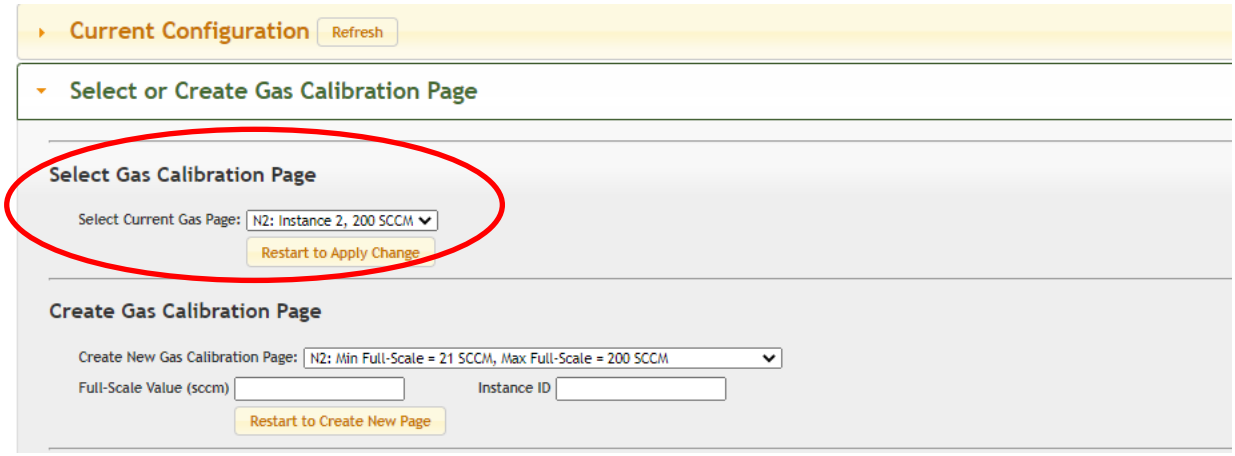


Figure 11: Current Gases Available

6.8.5.3 Wait for GFC to restart. A status box will open displaying “Restarting GFC…”. When the restart is complete, the box will go away. Please wait until this is complete.

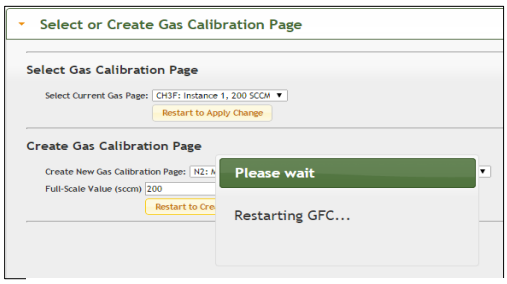


Figure 12: Saving New Process Gas

6.8.5.4 The gas change will be displayed in the Current Configuration Section. Check Current Gas Type and Full-Scale Flow settings to confirm.

6.8.6 How to add or update an existing Custom Gas list

6.8.6.1 Use the “Create New Gas Calibration Page” pull down to select the desired gas. Minimum and maximum allowed full scale values are listed next to the gas name.

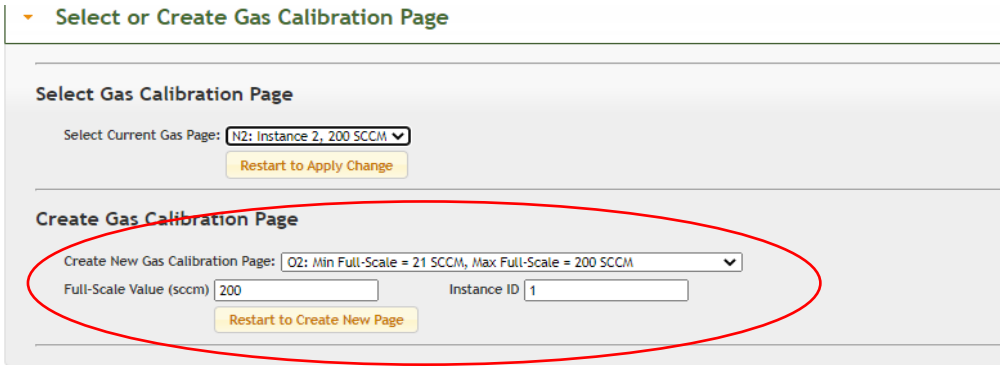


Figure 13: Adding New Gas

6.8.6.2 Enter the required full scale value in that field. Enter an instance ID in that field. Make sure the instance ID match the ID # in the tool (chamber).

Note:

For normal configurations use instance ID of 1 for process gas. Choosing an instance ID that already exists will overwrite the gas in that instance. Choose the next sequential number to add to the list. The software allows non-sequential instances (gaps in instance numbers is supported). If the communication protocol is DeviceNet, some OEM tools may not know how to process the pages when there are gaps between the instance numbers.

If the GFC will be flowing N2 into atmosphere pressures, N2_CAL_ATM must be selected as the gas.

6.8.6.3 Click the “Restart to Create New Page” button. A status box will open displaying “Restarting GFC…”. When the restart is complete, the box will go away. Check the “Select Current Gas Page”, the new created gas should be available for selection.

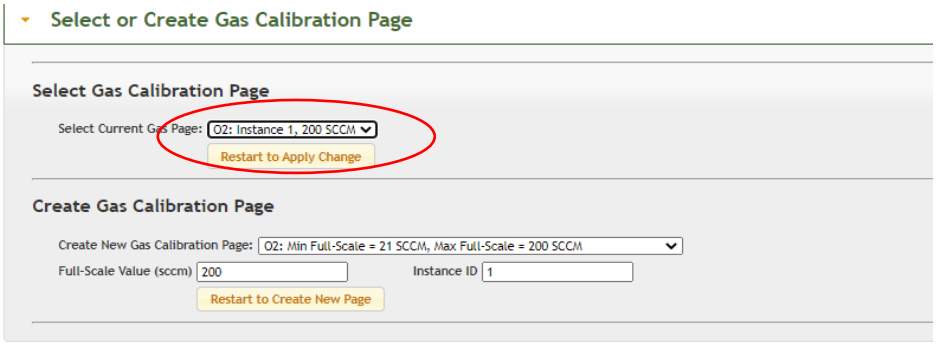


Figure 14: New Gas Created for Selection

6.8.7 How to change the Full Scale of the Gas in the Custom Gas list

If the tool requires a different maximum flow rate from the configured full-scale flow, the user may change it.

1. Open the “Update Full-Scale Flow” page, enter the new Full-Scale Flow rate desired. Maximum and minimum allowed Full-Scale Flow values for this gas and bin are noted to the right of the field.

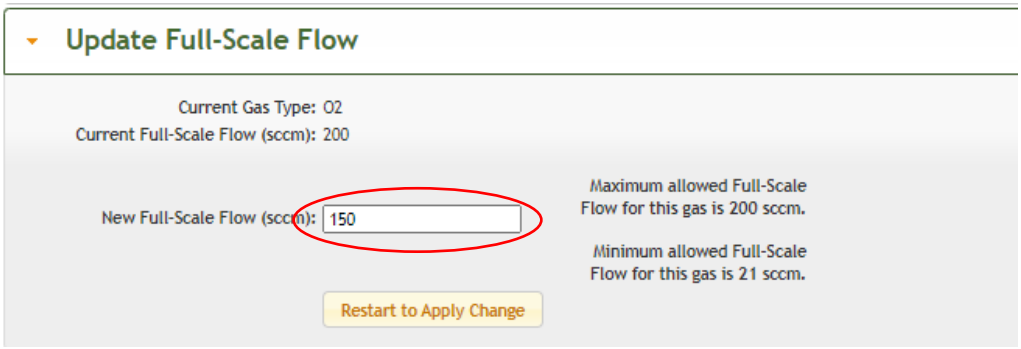


Figure 15: New Full Scale

2. Click “Restart to Apply Change”.
3. A message will pop up saying the GFC is restarting, please wait. When that message goes away, the new full scale can be confirmed on both the Current Configuration tab and the Gas Calibration tab.
4. If the selected full scale flow is larger than the device can flow, an error message will appear stating maximum full scale flow.

6.9 Calibration and Leak Check Utilities

This section addresses the function and use of the Calibration and Leak Check Utility section of the GFC User Interface. It does not identify the conditions under which calibration should be performed. Please refer to section 1 for full details and conditions required for calibration.

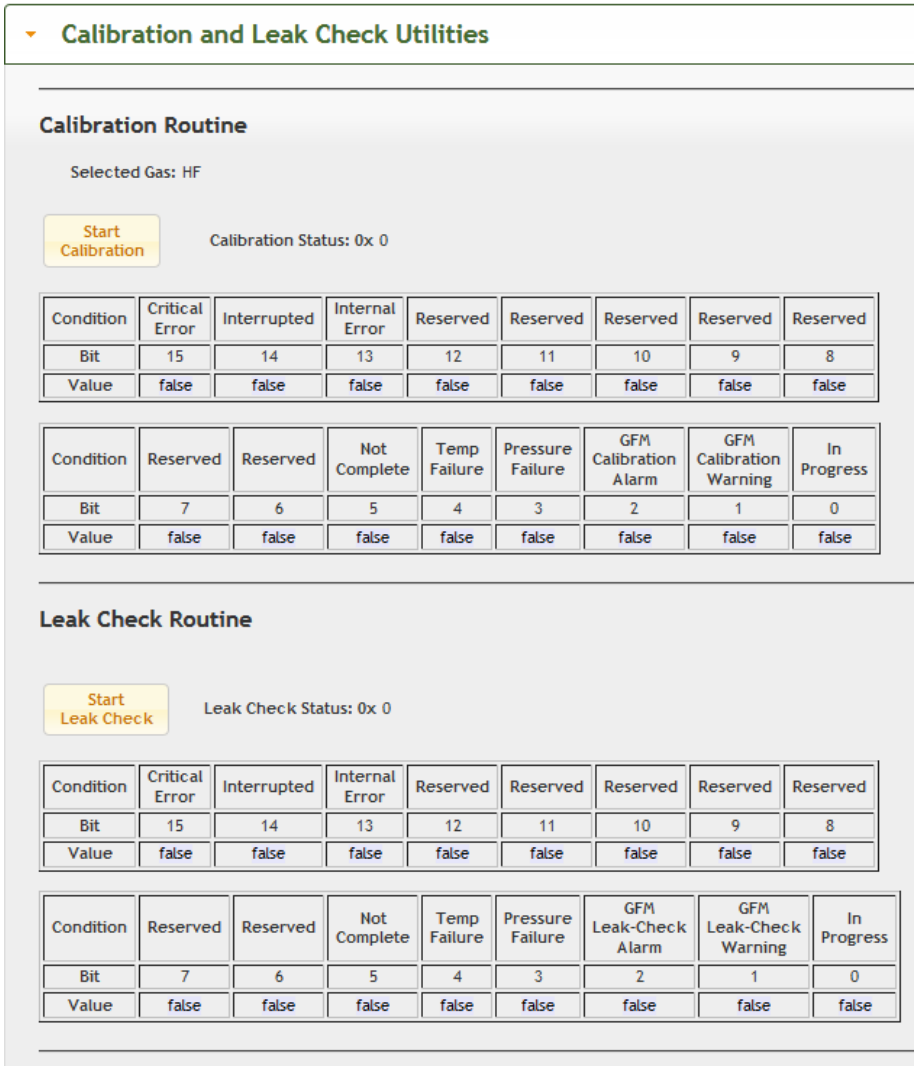


Figure 16: Cal-Leak Page

The Calibration and Leak Check Utility tab allows the user to either run the GFC Full Leak Check Routine, or the Calibration Routine, which performs the Full Leak Check prior to calibrating the GFC for use with the selected gas. Both sections provide Status of the routines as well as information on Warnings, Errors and Failures. Warnings indicate that values are moving out of optimal range. Warning flagged items should be monitored. Errors indicate values

outside of acceptable range. Immediate action should take place in the event of an Error. A Failure indicates a problem with reading a value or with the gas stick conditions.

6.9.1 Calibration Routine

The customer Tech/Engineer needs to open both the manual valve and the pneumatic valves to allow the gas to flow when the GFC is open. The GFC software will not control the gas panel valves.

The GFC Calibration Routine will first check to make sure that the appropriate pressure conditions exist to run the routine. If the conditions are met, it will check the GFM and GFC valves for internal leak rate. Finally, it will run a set of flows to calibrate the GFC.

6.9.1.1 Start Calibration

To initiate Calibration, click on Start Calibration. A pop-up will appear stating that if OK button is clicked, it will not be possible to view or modify configuration parameters for about 2 minutes. Click OK to start Calibration or Cancel to cancel calibration. When OK is selected, a box will appear saying "Please Wait, Calibrating…".; the Calibration status next to the Start Calibration button will change, and the "In Progress" field will turn blue and will read "true". When Calibration is successful, a window will pop up. Hit OK.

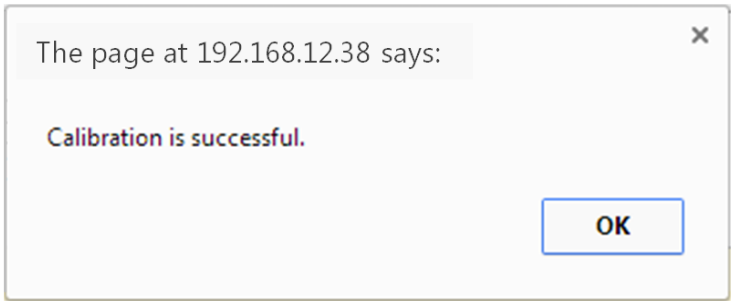


Figure 17: Cal Message

6.9.2 Calibration Failure

If the calibration fails, the software will display a pop-up window saying the program has failed. When OK button is clicked, the "Please Wait, Calibrating..." box will go away and the failure mode is displayed on the chart. The Not Complete field will be colored red and read "true".

6.9.2.1 Pressure Failure

A pressure failure will be displayed by a value of "true" and a red highlight in the Pressure Failure field. A Pressure Failure indicates that there was inadequate upstream or downstream pressure. In the event of a Pressure Failure, check to make sure that the upstream and downstream isolation valves are open, that there is adequate gas supply pressure upstream and adequate vacuum downstream.

6.9.2.2 Temperature Failure

The Calibration Routine checks the GFC temperature at the beginning of calibration and 30 seconds after calibration begins. If the temperature change is larger than allowed, calibration will fail. In this case, allow the GFC to come to a stable temperature before reinitiating the Calibration Routine.

6.9.2.3 Leak Check Warning

If the GFC detects a leak during the Calibration Routine, it will show as "true" in the Leak Check Routine section. The Calibration Routine may complete successfully, but if the GFC or GFM valves have a leak above the warning threshold, the Run LED will display a solid amber light. In this case, rerunning Calibration Routine is recommended. If Warning persists, monitor the GFC by routinely running Leak Check to ensure the leak has not grown.

6.9.2.4 Leak Check Error Failure

If the GFC detects an Error during the Calibration routine, the Calibration Routine will fail. The value will show as "true" in the Leak Check Routine section and the GFM or GFC Leak Error field on the Diagnostic Data tab will also show as "true". If the measured GFC leak is above Alarm threshold, the Run LED will display a blinking red light. If the measured GFM leak is above Alarm threshold, the Run LED will display a solid red light. If an error appears, rerunning Calibration routine is recommended. If the Error persists, replace the GFC.

6.9.3 Leak Check Routine (also known as Full Leak Check Routine)

The Leak Check Routine runs only the Leak Check portion of the Calibration Routine.

6.9.3.1 Run Leak Check Routine

To initiate the Leak Check Routine, click on Start Leak Check. A pop-up will appear stating that if OK button is clicked, it will not be possible to view or modify configuration parameters for about 2 minutes. Click OK to start Leak Check Routine or Cancel to cancel Leak Check Routine. When OK is selected, a box will appear saying "Please Wait, Performing Leak Check Routine…", the Leak Check Routine status next to the Start Leak Check button will change, and the In Progress field will turn blue and will read "true". When Leak Check Routine is successful, a window will pop up. Hit OK.

6.9.4 Leak Check Routine Failure

If the Leak Check Routine fails, the software will display a pop-up window saying the program has failed. When 'OK' button is clicked, the "Please Wait, Performing Leak Check Routine..." box will go away and the failure mode is displayed on the chart. The Not Complete field will be colored red and read "true".

6.9.4.1 Pressure Failure

A pressure failure will be displayed by a value of "true" and a red highlight in the Pressure Failure field. A Pressure Failure indicates that there was inadequate upstream or downstream

pressure. In the event of a Pressure failure, check to make sure that the upstream and downstream valves are open, that there is adequate gas supply pressure upstream and adequate vacuum downstream.

6.9.4.2 Temperature Failure

The Leak Check Routine checks the GFC temperature at the beginning of Leak Check and 30 seconds after the routine begins. If the temperature change is larger than allowed, Leak Check will fail. In this case, allow the GFC to come to a stable temperature before reinitiating the Leak Check routine.

6.9.4.3 Leak Check Warning

If the GFC detects a leak during the Leak Check routine, it will show as “true” in the Leak Check Routine section. If the GFC or GFM valves have a leak above the warning threshold, the Run LED will display a solid amber light. In this case, rerunning Calibration routine is recommended. If the Warning persists, monitor the GFC by routinely running Leak Check to ensure the leak has not grown.

6.9.4.4 Leak Check Error

If the GFC detects an Error during the Leak Check routine, the Leak Check Routine will fail. The value will show as “true” in the Leak Check Routine section and the GFM or GFC Leak Error field on the Diagnostic Data tab will also show as “true”. If the measured GFC leak is above Alarm threshold, the Run LED will display a blinking red light. If the measured GFM leak is above Alarm threshold, the Run LED will display a solid red light. If an error appears, rerunning Leak Check routine is recommended. If the Error persists, replace the GFC.

6.10 Diagnostic Data

The Diagnostic Data tab shows the current state of the Conditions listed on the chart. Note: if the GFC is not currently taking a GFM measurement, No GFM Flow Rate will indicate “true”.

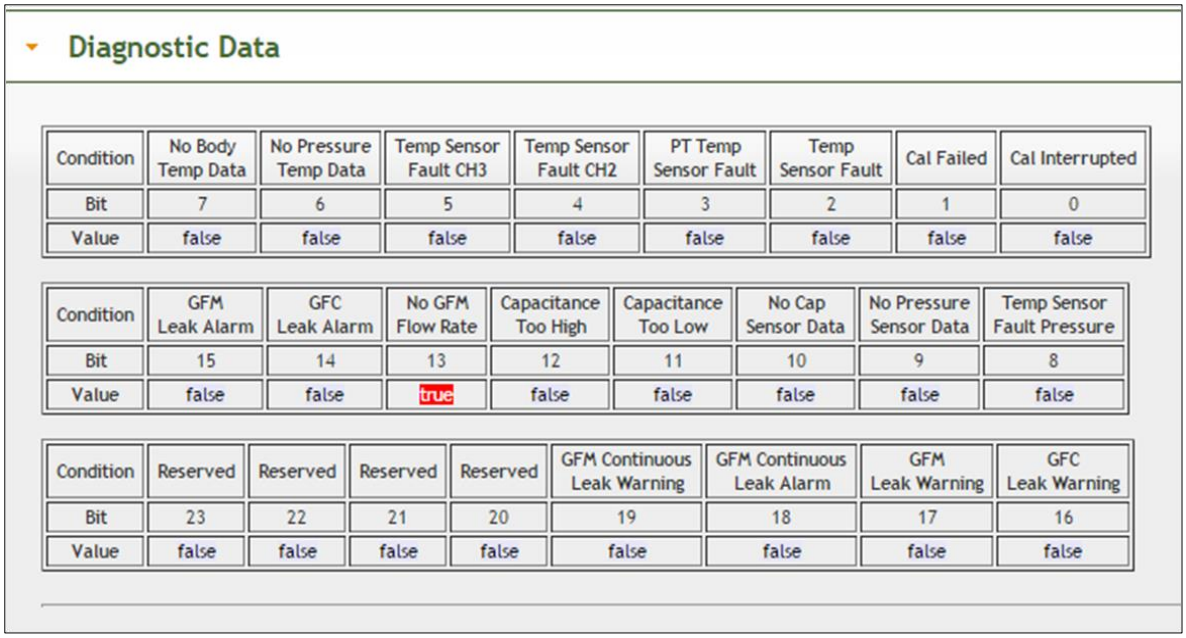


Figure 18: Diagnostic Data

6.11 Update RS-485 Configuration

This tab is used to check the status of the RS-485 MAC ID and Default Baud Rate. These values can be set using the switches on top of the GFC or can be set by the tool if the MSD and LDS switches are set to 0-0 (Program Mode). In Program Mode, these values can be set in the MAC ID tab, although any value set in this tab may be overwritten by the tool at any time. If the switches are not in Program Mode, any changes to the values on this tab will be overwritten by the switch values.

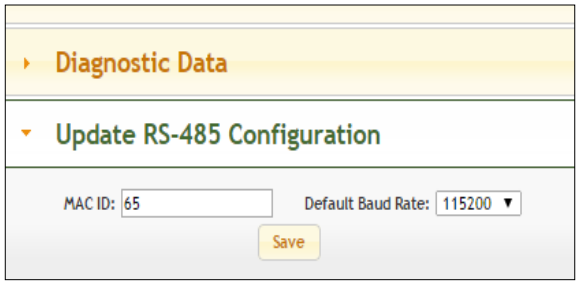


Figure 19: RS-485 Configuration

6.12 Update Device Net Configuration

This tab allows the change of the values shown. To make a change, use the dropdown menu on that value, and then hit the Save button on the bottom of the screen. Make sure the new value matches the tool (chamber) parameters. Supervisor Alarm Enable and Supervisor Warning Enable indicate that warnings and alarms will be enabled or disabled as indicated in the Sensor Configuration section. If Supervisor Alarms or Supervisor Warnings are disabled, all warnings or alarms are disabled. Changes can be verified by hitting the Refresh button on the Current Configuration tab header. The updated values will be shown on that tab. The unit does not need to be restarted to update the configuration.

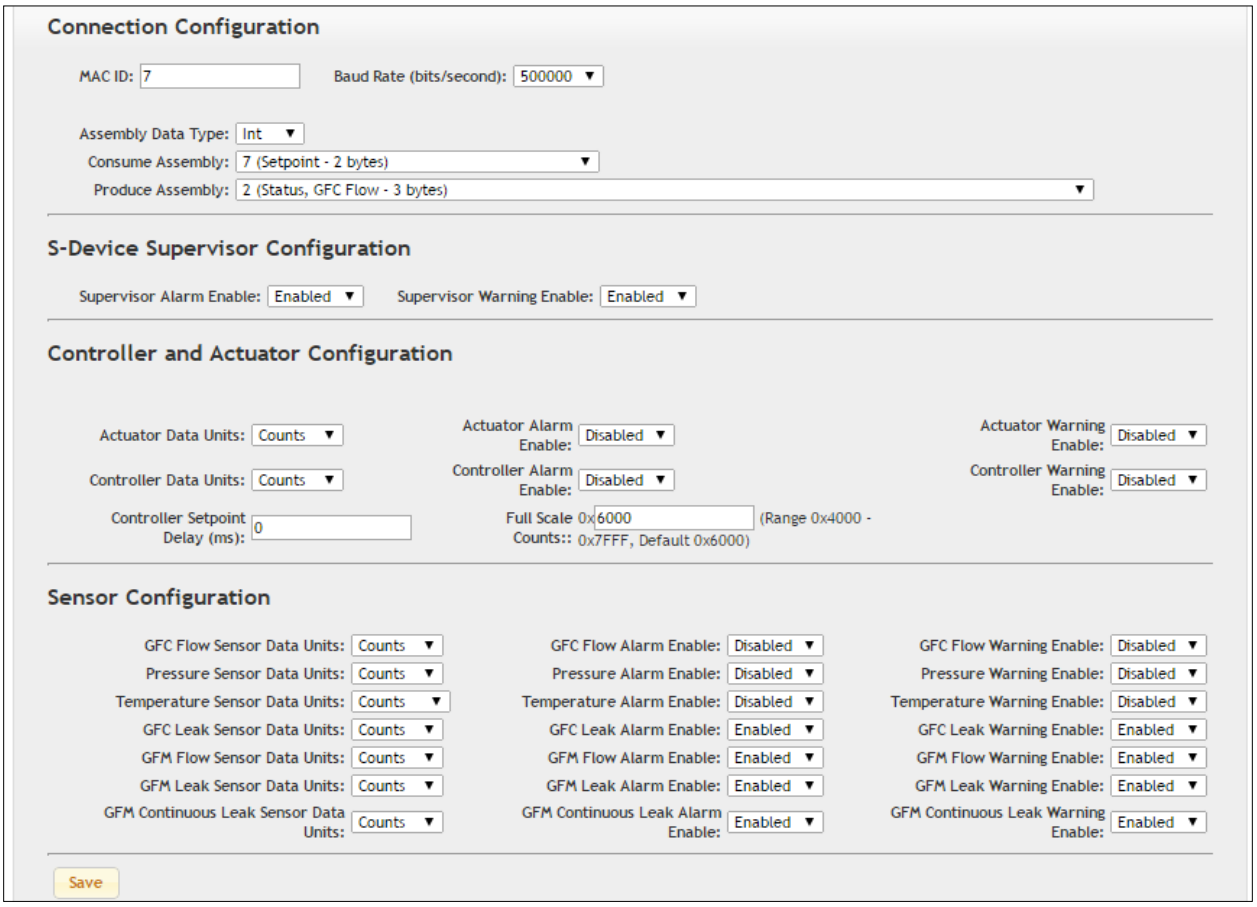


Figure 20: DNet Configuration Page

6.13 Update Ramping Recipe Configuration

Ramping Recipe Configuration settings are used to support tools that use very small setpoint changes to create a flow ramp. This tab is used to control how the GFC responds to the small setpoint changes. By default, this feature is enabled.

The GFC will treat any setpoint which is the lower of:

- 0.2% of full scale or
- Maximum Ramp Increment

as a ramp recipe. When GFC detects a ramp recipe condition, it will suspend flow correction activities from setpoint to setpoint to create a smooth ramp.

6.13.1 There are two Recipe Ramp fields that can be modified.

6.13.1.1 The first field allows Ramping Recipe to be enabled or disabled. Set Ramping Recipe State to 1 to enable this feature, or to 0 to disable this feature. The default value is 1.

6.13.1.2 The second field, Maximum Ramp Increment, allows the user to set the maximum ramp step size. By default, the Maximum Ramp Increment is limited by the 0.2% of Full Scale of the device. The software will display the value of 0.2% of full scale in the text describing the Maximum Ramp Increment. The default value of Maximum Ramp Increment is 100 msccm (0.1 sccm).

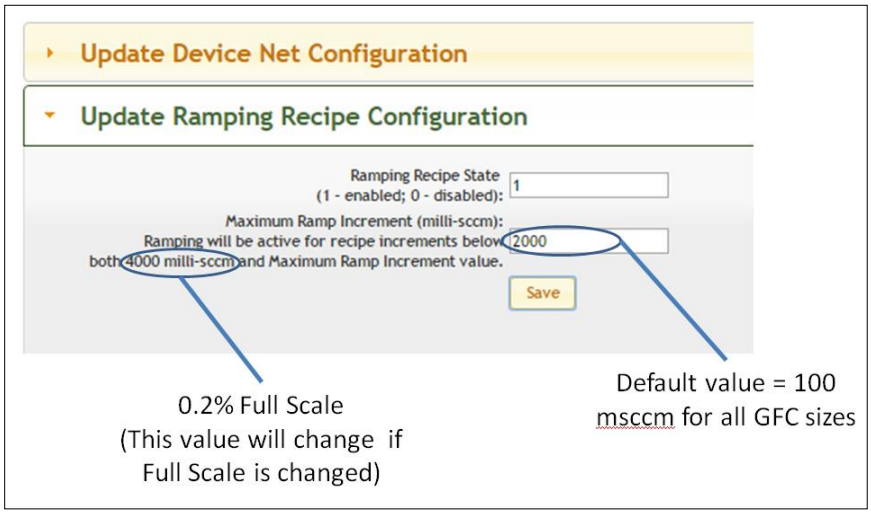


Figure 21: Recipe Configuration Page

6.13.1.3 Examples of triggering Recipe Ramp flows

GFC Unit Full Scale (SCCM)	0.2% of Full Scale (SCCM)	Max Ramp Increment (SCCM)	Largest Setpoint to trigger Ramp (SCCM)
2000* (Bin4)	**4**	**0.1**	**0.1**
2000	4	2	2
2000	4	6	4
700	1.4	6	1.4
1000* (Bin3)	**2**	**0.1**	**0.1**
1000	2	0.5	0.5
200* (Bin2)	**0.4**	**0.1**	**0.1**
200	0.4	0.2	0.2
200	0.4	1	0.4
55	0.11	0.2	0.11
20* (Bin1)	0.04	**0.1**	**0.04**
20	0.04	2	0.04
10	0.02	2	0.02
*** Default Settings**			

6.14 Update Flow Turn Off Configuration

Under this configuration, the user can choose:

Flow Turn Off Enable or Disable. Default Disable

Auto Stop Turn Off Enable or Disable. Default Disable

- Two turn off curves available
  - Default flat – continues set point
  - Optional slow turn off – Pivotal engineer configures slow turn off curve

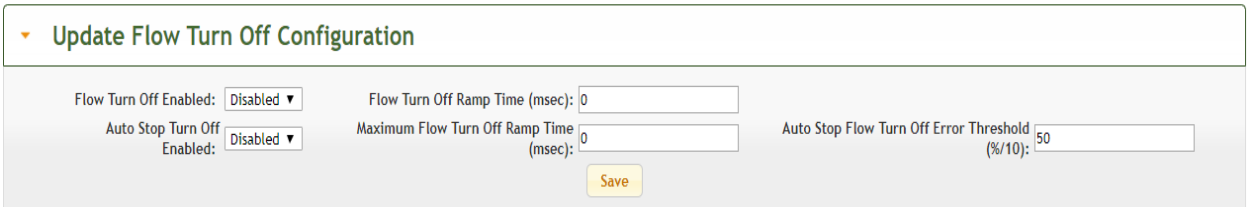


Figure 22: Flow Turn Off Page-Default Setting

6.15 Update Pressure Condition

Choose “Regular” from the dropdown menu for small pressure change during the flow in the Vacuum chamber. This applies to most chambers that are running low pressure processes.

Choose “High Downstream” for Downstream pressure changes from close to vacuum to close to atmosphere during the process flow.

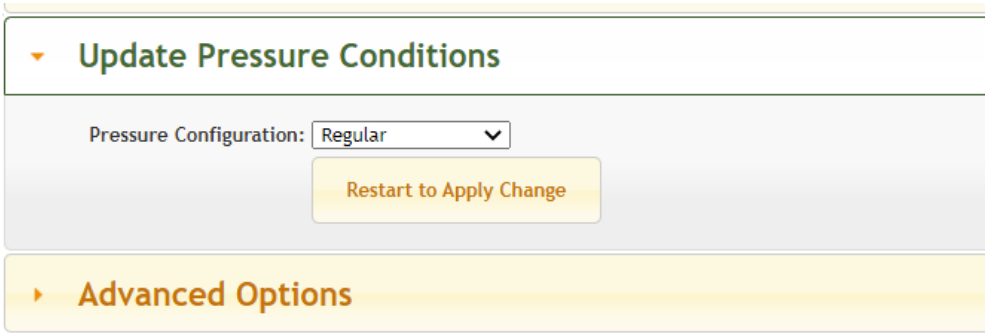


Figure 23: Pressure Condition Page

7 GFC Analog Adjustment

7.1 Scope

This procedure allows an end user to calibrate GFC analog input and output to the tool. This allows flexibility on installations where the GFC reading of analog setpoint and the GFC output of analog feedback does not correspond with what is supplied or read on a connected tool controller.

7.2 Definitions

GFC: The Pivotal Gas Flow Controller under calibration

Communication Mode: The current communication protocol the GFC is operating on

External Tool: The device sending an analog setpoint signal to the GFC and reading the analog feedback signal from the GFC

7.3 Safety Notice

WARNING!

Prior to any assembly, follow these important safety practices:

- Always wear gloves when working on or handling the GFC device.
- Always use protective plastic sheets or clean wipe sheets on the work surface to avoid scratching the GFC device and its components.
- Wear gloves and safety goggles when working with glues, epoxies, alcohol, and solvents.

7.4 Components

1) GFC to calibrate
2) IP connection to the GFC Ethernet port
3) MFC analog controller with setpoint input and output

7.5 Device Setup

1) With the GFC off and disconnected, configure the GFC for analog communications by setting the Data Rate switch to **A,** and the switch on the side of the GFC set to **DB9**.
2) Power the GFC and connect it to a local network via Ethernet cable
3) In a web browser that is also on this local network, navigate to 192.168.12.38/gfc/CustomerAnalogAdjustmentUI.html. Or under the CustomerUI, open the "Advanced Options", and click on "Analog Adjustment".

7.6 Front Panel

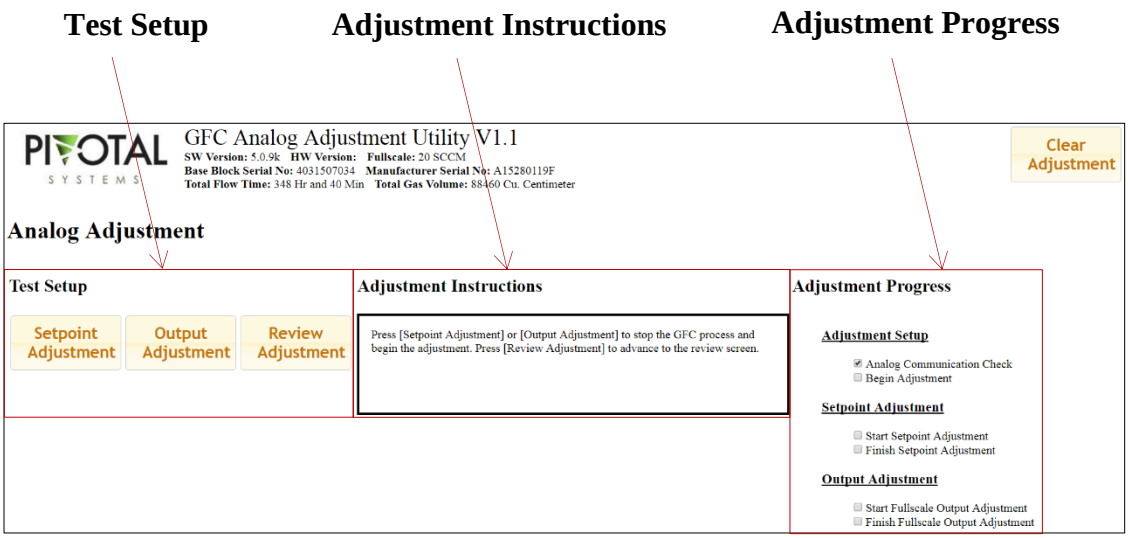


Figure 24: Analog Adjustment Main Screen

7.7 Adjustment Selections

7.7.1 Setpoint Adjustment

The Setpoint Adjustment reads a series of setpoints from the Tool and matches the voltage received with the tool command.

7.7.2 Output Adjustment

The Output Adjustment sends a series of flow feedback signals, user enters feedback read from tool and matches the GFC feedback voltage with the feedback read by the tool.

7.7.3 Review Adjustment

Review Adjustment provides the user to ensure that the commanded setpoint by the tool is calibrated with the GFC under the analog communication protocol. Under this menu, the Restart selection allows the user to recalibrate the adjustment for the Tool with the GFC.

7.7.4 Clear Adjustment

Clear Adjustment is used to remove the calibration.

7.7.5 Feature Selections

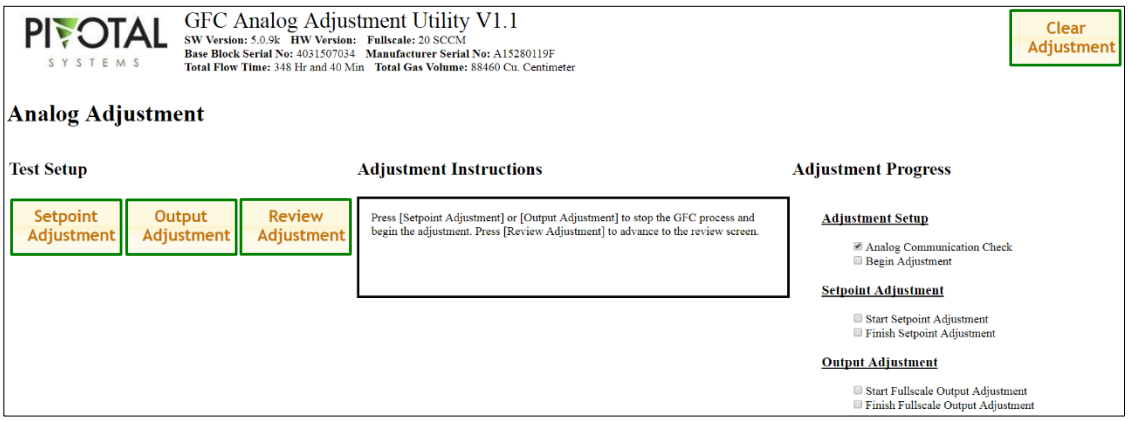


Figure 25: Feature Selection

7.8 Calibration

The Calibration Instructions window will guide you through the calibration sequence. The Calibration can be done using only the Input Adjustment or the Output Adjustment or both.

7.8.1 Input Adjustment

7.8.1.1 To begin calibrating the GFC to the Tool, select Setpoint Adjustment.

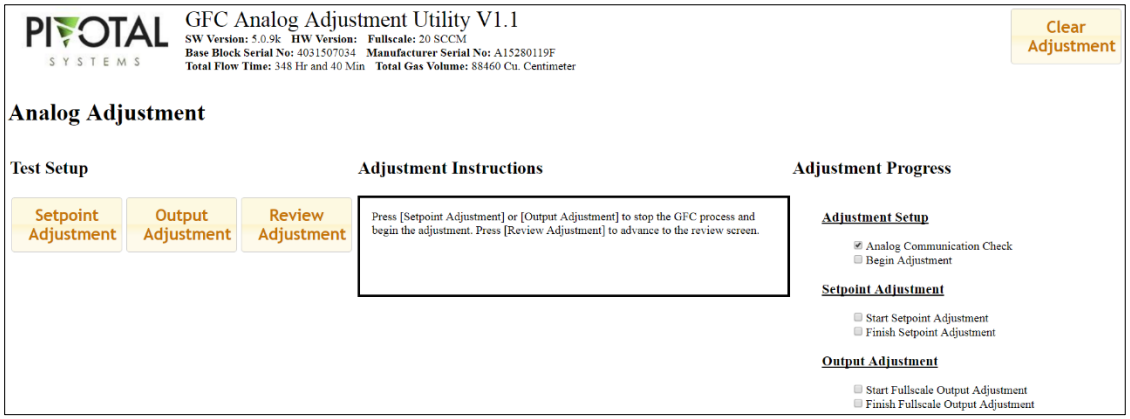


Figure 26: Setpoint Adjustment

7.8.1.2 Adjust the Tool setpoint control to the specified flow rate (10 % of full scale). This will be based on the calibrated Fullscale flow rate of the GFC. (GFC will not flow gas during this calibration)

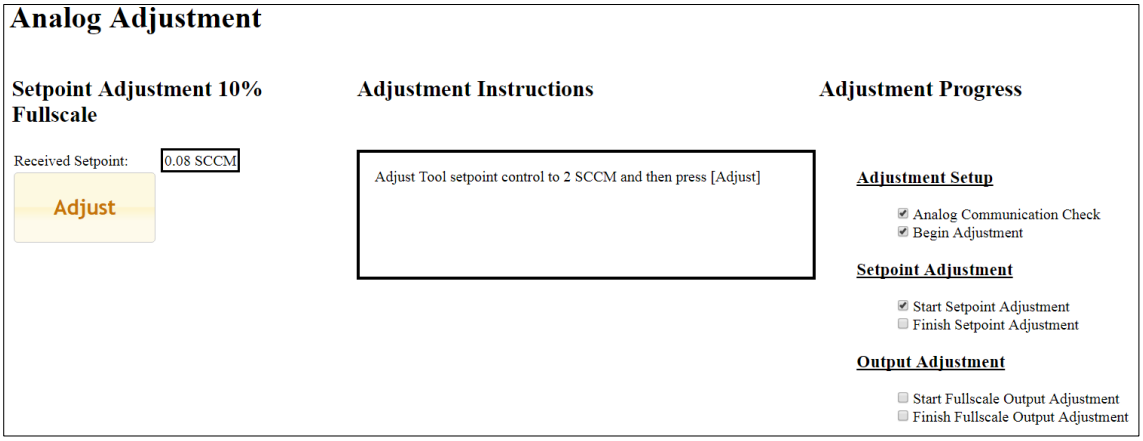


Figure 27: 10% Adjustment

7.8.1.3 Once the setpoint is received, select Adjust. This takes you to the next setpoint that is 25% of Fullscale. Repeat the previous step with the new setpoint defined in the Adjustment Instructions section.

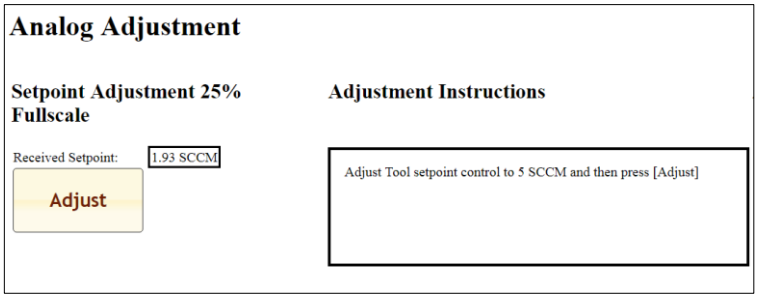


Figure 28: 25% Adjustment

7.8.1.4 Repeat previous step for 50% of Fullscale.

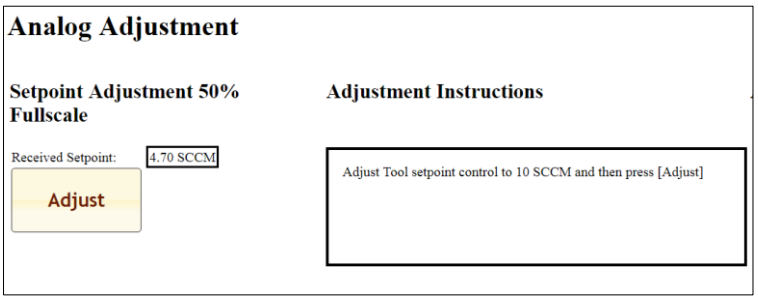


Figure 30: 50% Adjustment

7.8.1.5 Repeat previous step for 75% of Fullscale.

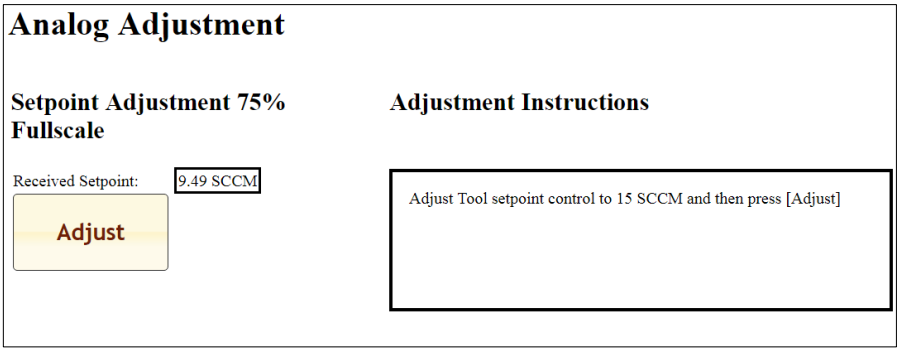


Figure 29: 75% Adjustment

7.8.1.6 Repeat previous step for 100% of Fullscale.

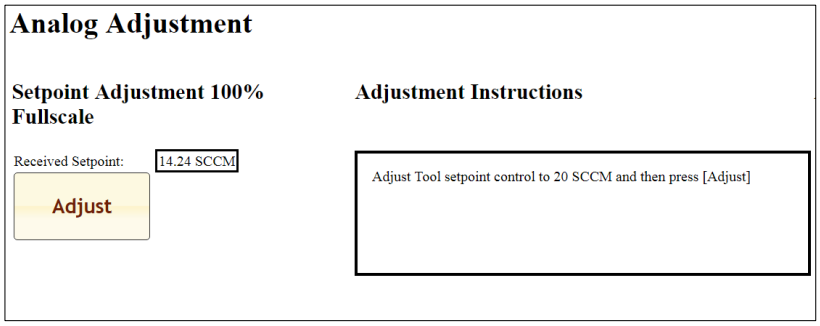


Figure 31: 100% Setpoint

7.8.1.7 The maximum absolute error will be reported in the pop-up window.

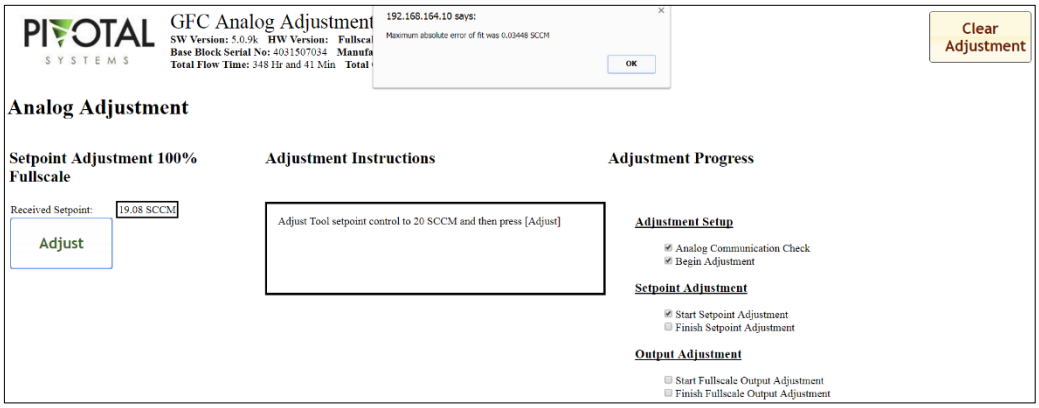


Figure 32: Maximum Error Message

7.8.2 Output Adjustment

7.8.2.1 The Output Adjustment can be set by selecting the Adjust Output.

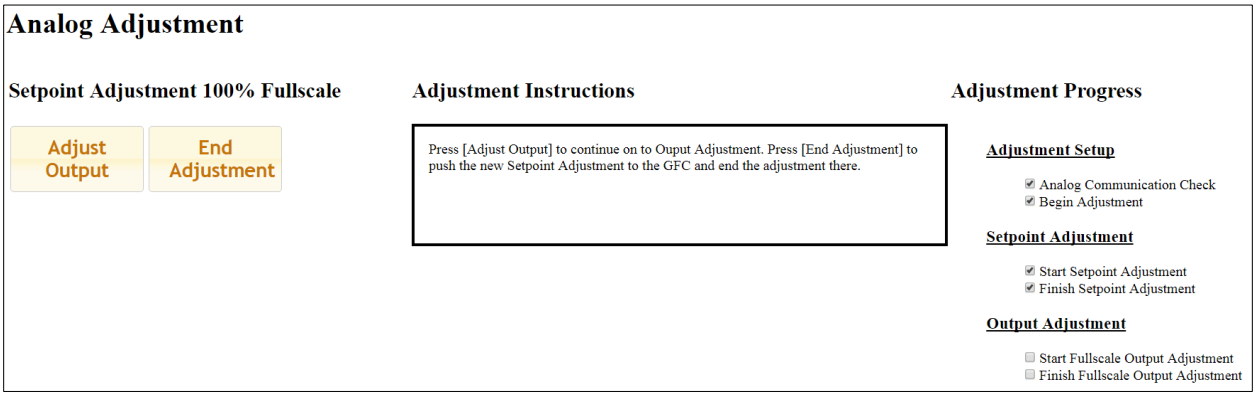


Figure 33: Maximum Error Message

7.8.2.2 Enter the flow reading from the Tool into the empty box under “Enter Tool Reading” and select Adjust.

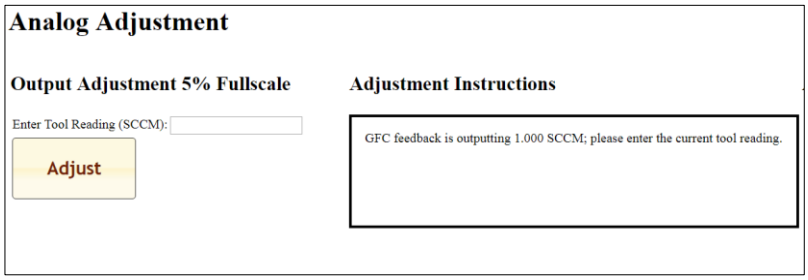


Figure 34: Enter 5% Reading

7.8.2.3 Repeat previous step for 20% of Fullscale and select Adjust.

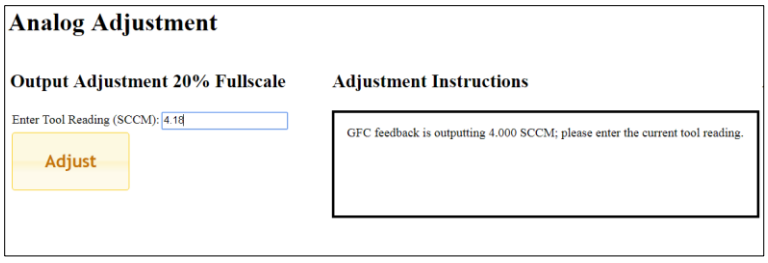


Figure 35: Enter 20% Reading

7.8.2.4 Repeat previous for 40% of Fullscale and select Adjust.

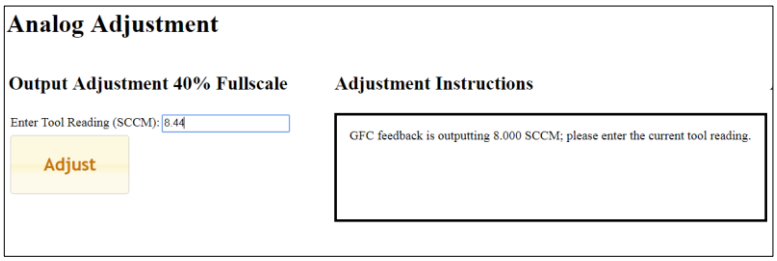


Figure 36: Enter 40% Reading

7.8.2.5 Repeat previous for 60% of Fullscale and select Adjust.

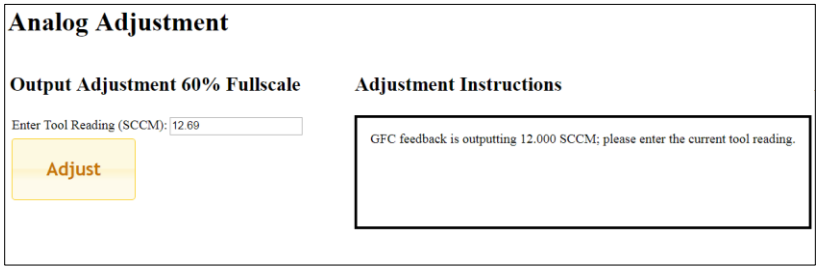


Figure 37: Enter 60% Reading

7.8.2.6 Repeat previous for 80% of Fullscale and select Adjust.

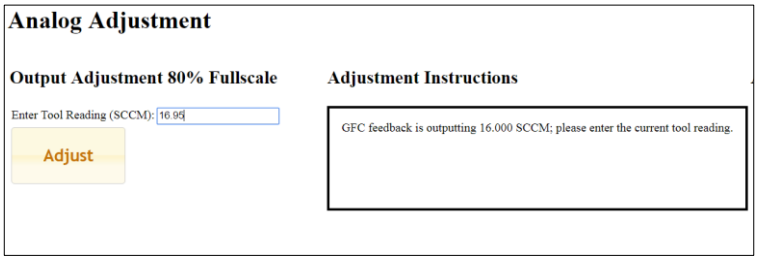


Figure 38: Enter 80% Reading

7.8.2.7 Then, the *maximum absolute error* will appear on a pop-up box.

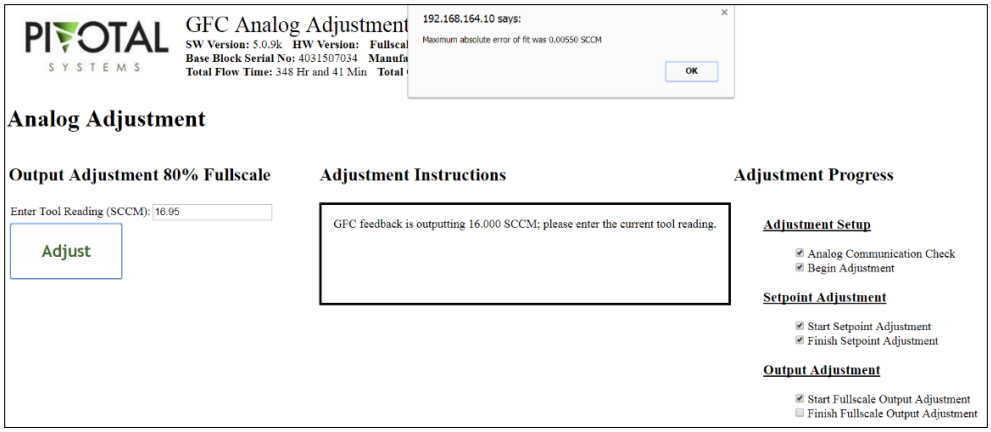


Figure 39: Maximum Error Message

7.8.2.8 Once the adjustment is complete, the program will report Adjustment Complete. Select Review Adjustment

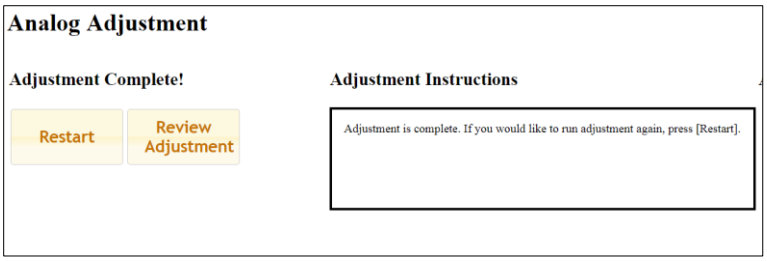


Figure 40: Adjustment Completion

7.8.2.9 After the adjustment is complete, the user can select **Review Adjustment** to test if the calibration was successfully adopted or **Restart** to restart the calibration process if results are not satisfactory. Close the web browser when the process is complete.

NOTE: If user selects Restart option, calibration process needs to be completed. Failure to do so can result in corrupted calibration.

To adjust another GFC, wait for 5 seconds, connect network cable to new GFC, and then open a new browser window to Adjustment url.

8 Communication Specifications

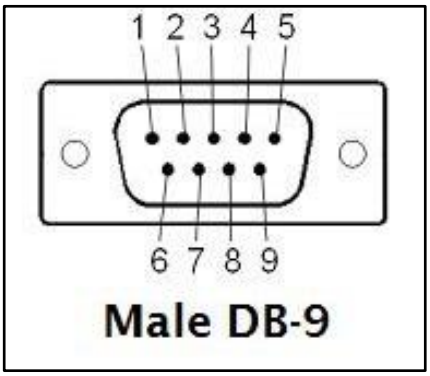
8.1 Analog Connection

Connect to the 9-pin Male D-sub connector on the top of the GFC enclosure when operating in the Analog mode. A suitable mating connector is Tyco PN# 1-747943-6.

Table 1: Allocation for Analog Connector Pins

Pin	Signal Name	Specification
1	Valve Override	+15VDC = valve fully opened -15VDC = valve forced closed GND/Open = no override
2	Flow Feedback	0 to 5VDC analog output 0 = no flow 5VDC = 100% flow
3	Positive Power Supply	+15VDC
4	Power Supply Common	Return for ±15VDC
5	Negative Power Supply	-15VDC
6	Flow Setpoint	0 to 5VDC analog input 0 = no flow 5VDC = 100% flow
7	Signal Ground	Return for pins 1, 2, and 6
8	*Positive RS485 Data*	*RS-485 serial data line*
9	*Negative RS485 Data*	*Return for pin 8*

Note: In the Analog mode, the GFC will ignore pins 8 and 9.



When Valve Override reaches +15VDC, the GFC valve will fully open to facilitate purging of the gas stick. Thus, the expected flow of a fully opened GFC valve can be much higher than a full-scale flow. The Flow Setpoint Analog Input has an input impedance of 20 kilohms, while the Flow Feedback Analog Output has a maximum output current of 30 milliamps.

The power requirement for the GFC when using the Analog mode of operation is as follows:

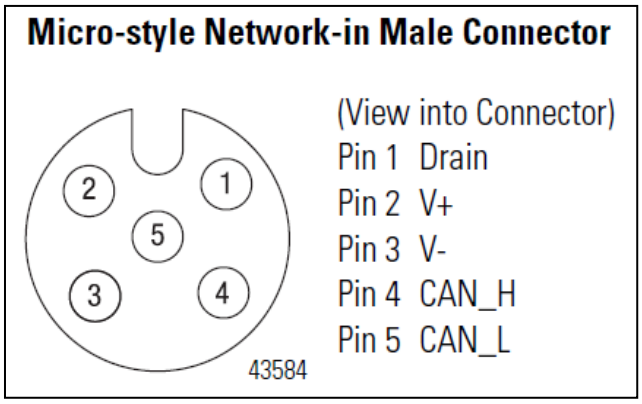
- +15VDC: 150mA continuous
- -15VDC: 150mA continuous

8.2 DeviceNet Connection

A standard DeviceNet "Thin" cable with "Micro" connector can connect to the side of the GFC enclosure.

Table 2: Connector Pins as per the ODVA Specification

Pin	Signal Name	Specification
1	Drain	
2	V+	+11VDC to +24VDC
3	V-	0VDC
4	CAN_H	
5	CAN_L	



The power requirement for the GFC when using the DeviceNet mode of operation is as follows:

- 700mA @ +11VDC
- 300mA @ +24VDC

8.3 RS-485 Connection

Connect to the 9-pin Male D-sub connector on the top of the GFC enclosure when operating in RS-485 mode. A suitable mating connector is Tyco PN# 1-747943-6.

Table 3: Allocation for RS-485 Connector Pin

Pin	Signal Name	Specification
1	Valve Override	+15VDC = valve fully opened -15VDC = valve forced closed GND/Open = no override
2	*Flow Feedback*	*0 to 5VDC analog output* *0 = no flow* *5VDC = 100% flow*
3	Positive Power Supply	+15VDC
4	Power Supply Common	Return for ±15VDC
5	Negative Power Supply	-15VDC
6	*Flow Setpoint*	*0 to 5VDC analog input* *0 = no flow* *5VDC = 100% flow*
7	Signal Ground	Return for pins 1, 2 and 6
8	Positive RS485 Data	RS-485 serial data line
9	Negative RS485 Data	Return for pin 8

Note: In the RS-485 mode, the GFC will ignore pins 2 and 6. However, it will support pin 1 (although an analog signal) in the RS-485 mode per the Lam Research 9-pin D-sub connector specification.

The power requirement for the GFC when using the RS-485 mode of operation is as follows:

- +15VDC: 150mA continuous
- -15VDC: 150mA continuous

9 Software Protocol

9.1 Analog Protocol and Signals

There are no software commands for the GFC while operating in Analog mode—the GFC is controlled through analog voltages only. Refer to section 8.1 for information on analog signals. Refer to 10.2.2 for Net and Mod LED status.

9.2 DeviceNet Protocol and Signals

The DeviceNet implementation adheres to the specifications for the ODVA CIP device profile for an Enhanced MFC (0x27) and conforms to the ODVA Semi-SIG DeviceNet interface guidelines.

Some common required objects include:

- Identity object
- Message router object
- DeviceNet object
- Connection object
- Connection manager object

In the following tables (with the exception of *Assembly Objects*), a decimal number in the first column denotes the row as an attribute ID, while a hexadecimal number denotes the row as a service number.

9.2.1 Assembly Objects

Table 4: Assembly Objects

Instance ID		Data Description	Data Type	Size (Bytes)
Consume Assembly (Input from the Wire to the Device)				
7	0x07	Setpoint	INT	2
8	0x08	Actuator Override, Setpoint	INT	3
19	0x13	Setpoint	REAL	4
20	0x14	Actuator Override, Setpoint	REAL	5
Standard Produce Assemblies (Output from the Device to the Wire)				

Instance ID		Data Description	Data Type	Size (Bytes)
1	0x01	GFC Flow	INT	2
2	0x02	Status, GFC Flow	INT	3
3	0x03	Status, GFC Flow, Valve	INT	5
4	0x04	Status, GFC Flow, Setpoint	INT	5
5	0x05	Status, GFC Flow, Setpoint, Valve	INT	7
6	0x06	Status, GFC Flow, Setpoint, Actuator Override, Valve	INT	8
21	0x15	Status, GFC Flow (1), Pressure (2), Temperature (3)	INT	7
22	0x16	Status, GFC Flow (1), Valve, Pressure (2), Temperature (3)	INT	9
13	0x0D	GFC Flow	REAL	4
14	0x0E	Status, GFC Flow	REAL	5
15	0x0F	Status, GFC Flow, Valve	REAL	9
16	0x10	Status, GFC Flow, Setpoint	REAL	9
17	0x11	Status, GFC Flow, Setpoint, Valve	REAL	13
18	0x12	Status, GFC Flow, Setpoint, Actuator Override, Valve	REAL	14
23	0x17	FP – Status, GFC Flow, Pressure and Temperature	REAL	13
24	0x18	FP – Status, GFC Flow, Valve, Pressure and Temperature	REAL	17
Manufacturer Specific Produce Assemblies (Output from the Device to the Wire)				
100	0x64	GFC Flow, GFM Flow	INT	4
101	0x65	Status, GFC Flow, GFM Flow	INT	5
102	0x66	Status, GFC Flow, Setpoint, Valve (Drive), GFM Flow	INT	9
103	0x67	Status, GFC Flow, Pressure, Temperature, GFM Flow	INT	9
104	0x68	Status, GFC Flow, Valve (Drive), Pressure, Temperature, GFM Flow	INT	11
105	0x69	Status, GFC Flow, Pressure, Temperature, GFM Leak	INT	9

Instance ID		Data Description	Data Type	Size (Bytes)
106	0x6A	Status, GFC Flow, Valve (Drive), Pressure, Temperature, GFC Leak, GFM Flow, GFM Leak	INT	15
112	0x70	Status, GFC Flow, Pressure, Temperature, Continues GFM Leak	INT	9
108	0x6C	Status, GFC Flow, Valve (Drive), Valve Position	INT	7
109	0x6D	Status, GFC Flow, Setpoint, Valve (Drive), Valve Position, GFM Flow	INT	11
110	0x6E	Status, GFC Flow, Valve (Drive), Valve Position, Pressure, Temperature, GFM Flow	INT	13
111	0x6F	Status, GFC Flow, Valve (Drive), Valve Position, Pressure, Temperature, GFC Leak, GFM Flow, GFM Leak	INT	17
120	0x78	GFC Flow, GFM Flow	REAL	8
121	0x79	Status, GFC Flow, GFM Flow	REAL	9
122	0x7A	Status, GFC Flow, Setpoint, Valve (Drive), GFM Flow	REAL	17
123	0x7B	Status, GFC Flow, Pressure, Temperature, GFM Flow	REAL	17
124	0x7C	Status, GFC Flow, Valve (Drive), Pressure, Temperature, GFM Flow	REAL	21
125	0x7D	Status, GFC Flow, Pressure, Temperature, GFM Leak	REAL	17
126	0x7E	Status, GFC Flow, Valve (Drive), Pressure, Temperature, GFC Leak, GFM Flow, GFM Leak	REAL	29
132	0x7E	Status, GFC Flow, Pressure, Temperature, Continues GFM Leak	REAL	17
128	0x80	Status, GFC Flow, Valve (Drive), Valve Position	REAL	13
129	0x81	Status, GFC Flow, Setpoint, Valve (Drive), Valve Position, GFM Flow	REAL	21
130	0x82	Status, GFC Flow, Valve (Drive), Valve Position, Pressure, Temperature, GFM Flow	REAL	25
131	0x83	Status, GFC Flow, Valve (Drive), Valve Position, Pressure, Temperature, GFC Leak, GFM Flow, GFM Leak	REAL	33

9.2.2 S-Device Supervisor (0x30), Instance 1

Table 5: S-Device Supervisor

Attribute ID or Service Number (DEC/HEX)	Name	Size	Data Type
1 (Class Attribute)	Revision	2	UINT
3	Device Type	9 (8 chars)	SHORT STRING
4	SEMI Standard Revision Level	9 (8 chars)	SHORT STRING
5	Manufacturer's Name	21 (20 chars)	SHORT STRING
6	Manufacturer's Model Number	21 (20 chars)	SHORT STRING
7	Software Revision Level	7 (6 chars)	SHORT STRING
8	Hardware Revision Level	7 (6 chars)	SHORT STRING
9	Manufacturer's Serial Number	31 (30 chars)	SHORT STRING
11	Device Status	1	USINT
12	Exception Status	1	BYTE
13	Exception Detail Alarm	8	Array of Exception Details
14	Exception Detail Warning	8	Array of Exception Details
15	Alarm Enable	1	BOOL
16	Warning Enable	1	BOOL
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A
0x05	Reset	-	N/A
0x06	Start	-	N/A
0x07	Stop	-	N/A
0x4B	Abort	-	N/A
0x4C	Recover	-	N/A
0x4E	Perform Diagnostics	-	N/A

Table 6: Exception Status Byte Information (Attribute 12)

Bit	Function
0	ALARM / device-common
1	ALARM / device-specific
2	ALARM / manufacturer-specific (== 1 if GFM Leak Alarm, see details)
3	0 == Reserved
4	WARNING / device-common
5	WARNING / device-specific (== 1 if Not Reading Valid – S-Analog Sensor)
6	WARNING / manufacturer-specific (== 1 if GFM Leak Warning)
7	1 == Expanded Method

Table 7: Alarm Exception Detail (Bits 0, 1, or 2 of Exception Status set to High) (Attribute 13)

Device Common Section of Alarm Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Common Exception Size	0	0	0	0	0	0	1	0
Common Exception	Reserved	Internal Real-Time	Reserved	RAM	EEPROM	EEPROM	Micro processor	Internal Diagnostic
Common Exception	Reserved	Reset	Notify Manufacturer	Scheduled Maintenance Due	Power Supply Input Voltage	Power Supply Output Voltage	Reserved	Power Supply Over Current
Device Specific Section of Alarm Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Size	0	0	0	0	0	0	1	0
Specific Exception	Reserved	Reserved	Valve High S-Analog Actuator	Valve Low S-Analog Actuator	Flow Control S-Single Stage Controller	Flow High S-Analog Sensor	Flow Low S-Analog Sensor	Reserved
Specific Exception	Reserved	Reserved	Reserved	Reserved	Temperature High S-Analog Sensor (Instance 3)	Temperature Low S-Analog Sensor (Instance 3)	Pressure High S-Analog Sensor (Instance 2)	Pressure Low S-Analog Sensor (Instance 2)
Manufacturer Section for Alarm Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Size	0	0	0	0	0	0	0	1
Exception	0	Continuous GFM Leak High					GFM Leak High	GFC Leak High

Table 8: Warning Exception Detail (Bits 4, 5, or 6 of Exception Status set High) (Attribute 14)

Device Common Section of Warning Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Size	0	0	0	0	0	0	1	0
Common Exception	Reserved	Internal Real-Time	Reserved	RAM	EEPROM	EEPROM	Micro processor	Internal Diagnostic
Common Exception	Reserved	Reset	Notify Manufacturer	Scheduled Maintenance Due	Power Supply Input Voltage	Power Supply Output Voltage	Reserved	Power Supply Over Current
Device Specific Section of Warning Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Size	0	0	0	0	0	0	1	0
Specific Exception	Reserved	Reserved	Valve High S-Analog Actuator	Valve Low S-Analog Actuator	Flow Control S-Single Stage Controller	Flow High S-Analog Sensor	Flow Low S-Analog Sensor	Not Reading Valid S-Analog Sensor (Instance 1)
Specific Exception	Reserved	Reserved	Not Reading Valid S-Analog Sensor (Instance 3)	Not Reading Valid S-Analog Sensor (Instance 2)	Temperature High S-Analog Sensor (Instance 3)	Temperature Low S-Analog Sensor (Instance 3)	Pressure High S-Analog Sensor (Instance 2)	Pressure Low S-Analog Sensor (Instance 2)
Manufacturer Section for Warning Exception Detail								
Data Component	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Detail Size	0	0	0	0	0	0	0	1
Detail [0]	0	Continuous GFM Leak High				Calibration Routine Failed	GFM Leak High	GFC Leak High

9.2.3 S-Analog Sensor (0x31), 3 Standard Instances and 4 Manufacturer specific Instances

Table 9: **Instance 1: Flow Feedback**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
17	Alarm Trip Point High	2/4	INT or REAL
18	Alarm Trip Point Low	2/4	INT or REAL
20	Alarm Settling Time	2	UINT
21	Warning Trip Point High	2/4	INT or REAL
22	Warning Trip Point Low	2/4	INT or REAL
24	Warning Settling Time	2	UINT
28	Auto-zero Status	1	BOOL
35	Gas Calibration Object Instance	2	UINT
95	Flow Totalizer	8	ULINT
96	Flow Hours	4	UDINT
99	Subclass	2	UINT
100	Health Check One	2	UINT
101	Health Check Two	2	UINT
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A
0x4B	Zero Adjust	-	N/A

Table 10: **Instance 2: Pressure**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
12	Offset-A	2	INT
17	Alarm Trip Point High	2/4	INT or REAL
18	Alarm Trip Point Low	2/4	INT or REAL
20	Alarm Settling Time	2	UINT
21	Warning Trip Point High	2/4	INT or REAL
22	Warning Trip Point Low	2/4	INT or REAL
24	Warning Settling Time	2	UINT
28	Auto-zero Status	1	BOOL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A
0x4B	Zero Adjust	-	N/A

Table 11: **Instance 3: Temperature**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
12	Offset-A	2	INT
17	Alarm Trip Point High	2/4	INT or REAL
18	Alarm Trip Point Low	2/4	INT or REAL
20	Alarm Settling Time	2	UINT
21	Warning Trip Point High	2/4	INT or REAL
22	Warning Trip Point Low	2/4	INT or REAL
24	Warning Settling Time	2	UINT
28	Auto-zero Status	1	BOOL
103	Body Temperature Delta Alarm Threshold (in degrees)	2/4	INT or REAL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

Table 12: **Instance 100: GFC Leak**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

Table 13: **Instance 101: GFM Flow**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

Table 14: **Instance 102: GFM Leak**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
12	Offset-A	2	INT
28	Auto-zero Status	1	BOOL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A
0x4B	Zero Adjust	-	N/A

Table 15: **Instance 103: Continuous GFM Leak**

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Reading Valid	1	BOOL
6	Value	2/4	INT or REAL
7	Status	1	BYTE
10	Full Scale	2/4	INT or REAL
110	Full Scale		Struct
	Amount	4	REAL
	Units	2	ENG UNITS
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

9.2.4 S-Analog Actuator (0x32), 1 Instance

Table 16: S-Analog Actuator

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
5	Override	1	USINT
6	Value	2/4	*Data Type*
7	Status	1	BYTE
8	Alarm Enable	1	BOOL
9	Warning Enable	1	BOOL
15	Alarm Trip Point High	2/4	INT or REAL
16	Alarm Trip Point Low	2/4	INT or REAL
18	Warning Trip Point High	2/4	INT or REAL
19	Warning Trip Point Low	2/4	INT or REAL
21	Safe State	1	USINT
22	Safe Value	2/4	INT or REAL
100	Valve Position Sensor Adjust Status	2	WORD (see "Leak Check and Calibration Status" table below)
101	Valve Check Status	2	WORD (see "Leak Check and Calibration Status" table below)
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A
0x32	Valve Position Sensor Adjust	-	N/A
0x33	Valve Check	-	N/A

Table 17: Leak Check and Calibration Status

Leak Check and Calibration Status																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Condition	Critical Error	Interrupted	Internal Error								Not Complete	Temp Failure	Pressure Failure	GFM Leak Alarm	GFM Leak Warning	In Progress

9.2.5 S-Single Stage Controller (0x33), 1 Instance

Table 18: S-Single Stage Controller

Attribute ID or Service Number (DEC/HEX)	Name	Size (Bytes)	Data Type
3 (Attribute ID)	Data Type	1	USINT
4	Data Units	2	ENGUNITS
6	Setpoint	2/4	INT or REAL
7	Process Variable	2/4	INT or REAL
10	Status	1	BYTE
11	Alarm Enable	1	BOOL
12	Warning Enable	1	BOOL
13	Alarm Settling Time	2	UINT
14	Alarm Error Band	2/4	INT or REAL
15	Warning Settling Time	2	UINT
16	Warning Error Band	2/4	INT or REAL
19	Ramp Rate	4	UDINT
104	Ramp Rate NV/V Ram Storage	1	BOOL
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

9.2.6 S-Gas Calibration (0x34)

Table 19: Instance 0: General Information

Attribute ID or Service Number (DEC/HEX)	Name	Size	Data Type	
1 (Attribute ID)	Revision	2	UINT	
170	Bin ID	2	INT	
171	Revision	2	UINT	
0x32 (Service)	Create	12 + sizeof(SHORT_STRING) + sizeof(SHORT_STRING)	UINT + UINT + UINT + REAL + ENGUNITS + SHORT_STRING+ SHORT_STRING	Instance Gas Number Valid Instance Full-Scale F-S Data Units Gas Symbol Customer Gas Symbol
0x0E	Get Attribute Single	-	N/A	
0x4B	Get All Instances	2 + (6 * (Number of Instances))	UINT + Array of Struct{ UINT + UINT + UINT }	Number of Instances S-Gas Calibration Instance ID Gas Standard Number Valid S-Analog Sensor Instance ID

Table 20: Instance 1- N*, 1 Instance per Gas Type

Note: * - Gas Instance Number, Maximum number of instances is 255.

Note: The Instance Numbers are consecutive but have gaps between certain sets.

Attribute ID or Service Number (DEC/HEX)	Name	Size	Data Type
3 (Attribute ID)	Gas Standard Number	2	UINT
4	Valid Sensor Instance	2	UINT
5	Gas Symbol	32 (31 char)	SHORT STRING
6	Full Scale	-	Struct
	Amount	4	REAL
	Units	2	ENG UNITS
8	Calibration Date	2	DATE
9	Calibration Gas Number	2	UINT
95	Calibration Pressure	4	REAL
96	Calibration Temperature	4	REAL
99	Subclass	2	UINT
0x0E (Service)	Get Attribute Single	-	N/A
0x10	Set Attribute Single	-	N/A

9.3 DeviceNet Indicator Lights

9.3.1 LED Power-On Self-Test (POST)

Upon power-on, the device will perform a Power-On Self-Test on both the Network Status and Module Status LED*s*. While the device tests these LED*s* and their supporting hardware, it will perform the following LED color change sequence:

- Network Status and Module Status LED*s* are **off**.
- Turn Module Status LED on **green** for approximately 0.25 seconds.
- Turn Module Status LED on **red** for approximately 0.25 seconds.
- Turn Module Status LED on **green**.
- Turn Network Status LED on **green** for approximately 0.25 seconds.
- Turn Network Status LED on **red** for approximately 0.25 seconds.
- Network Status and Module Status LED*s* are **off**.

Then, the device goes to normal operation where LEDs will reflect the condition of the DeviceNet network.

9.3.2 Module Status (MOD) LED States

The Module Status (MOD) LED indicates the status of the underlying flow controller application.

Device State	LED State	Description
Not Powered	Off	No power applied to device
Self-Test	Flashing Green-Red	Device is executing a series of self-tests to ensure it is capable of transitioning to an operational state. In case of successful Self-Test light will transition to Solid Green
Self-Test Exception	Flashing Green-Red	If Self-Test was not successful device remains in this condition for an infinite amount of time
Idle or Executing	Solid Green (Normal Operation)	Idle = A safe state where the device has been initialized and successfully completed self-testing; it is not executing any operational components of an MFC Executing = The device is performing MFC-related tasks
Abort	Flashing Red	Device detected a recoverable fault

Critical Fault	Solid Red	Device detected an unrecoverable fault

Note: The *DeviceNet Specification* defines blink rates and patterns for the Module Status LED.

9.3.3 Network Status (NET) LED States

The Network Status (NET) LED indicates the status of the DeviceNet communication link.

Network State	LED State	Description
Not Powered or Not On-Line	Off	No power applied to the device or the device has not completed its Duplicate MAC ID check (not on-line). This condition may occur when device is only (lonely) node (no Master Controller or any other Device) on the network
On-Line and Not Connected	Flashing Green	Device is on-line, but no connections are in the Established state
Link OK	Solid Green	Device is on-line, and one or more connections are in the Established state
Connection Time-Out	Flashing Red	One or more connections timed out
Critical Link Failure	Solid Red	Device failed its Duplicate MAC ID check or the device detected a Bus-Off condition*

* Note: The device is now incapable of communicating on the network.

9.4 RS-485 Protocol and Signals

The GFC complies with RS-485 communication protocols as defined by FNET and Lam Research, Inc. Refer to section 10.2.1 for Net and Mod LED status.

10 Troubleshooting LED Lights

10.1 About GFC LEDs

The GFC device has several LED lights. The following sections decipher these lights and how to troubleshoot them, if applicable.

10.2 Power States (PWR LED)

The table below defines the GFC power conditions.

Power (PWR) LED Color	Behavior	Troubleshooting Comments
OFF	No Power to the unit or Possible LED Failure if other LEDs are ON	Check PWR switch position to make sure the selected position corresponds to the available power/communication source Check power supply Check power cable
Solid Green	**Normal State:** Power to the unit is good	
Faint Green	Error State: Power to the unit is low	Check power supply to unit

10.2.1 DB9-RS485 Power

This state is selected when RS-485 power is supplied through the DB-9 connector on top of the device. The COMM (PWR) selector switch must be set to **DB9**, and the Data Rate rotary switch must be set to **1, 2,** or **5**.

NET LED Color	Behavior	Troubleshooting Comments
Flashing Amber	Indicates RS-485 communication mode	If no communication, check MAC ID switches. If switch is between detents, it will not read the correct MAC ID and will not communicate. Note that when switch is in detent, it may not line up perfectly with silkscreen on case.
Solid Amber	Error State	Check Data Rate switch position to make sure it is in position **1**, **2**, or **5**. . If switch is between detents, it will not read the correct Data Rate and will not communicate. Note that when switch is in detent, it may not line up perfectly with silkscreen on case.

		If the LED is still flashing, contact a field service engineer or return the unit
Solid or Flashing Green/Red	Possible LED failure if the unit is communicating	Check if RS-485 communication is available
OFF	Possible LED failure if the unit is communicating	Check if RS-485 communication is available
MOD LED Color	**Behavior**	**Troubleshooting Comments**
OFF	Not in use in RS-485 communication mode	

10.2.2 DB9-Analog Power

This state is selected when RS-485 power is supplied through the DB-9 connector on top of the device. The COMM (PWR) selector switch must be set to **DB9**, and the Data Rate switch must be set to **A**.

NET LED Color	Behavior	Troubleshooting Comments
Solid Amber	Indicates Analog communication mode	
Flashing Amber	Error State	Check Rate switch position to make sure it is in the A. If switch is between detents, it will not read the correct Data Rate and will not communicate. Note that when switch is in detent, it may not line up perfectly with silkscreen on case. If the LED is still flashing, contact a field service engineer or return the unit
Solid or Flashing Green/Red	Possible LED failure if the unit is communicating	Check if Analog communication is functional
OFF	Possible LED Failure if the unit is communicating	Check if Analog communication is functional
MOD LED Color	**Behavior**	**Troubleshooting Comments**
OFF	Not in use in Analog communication mode	

10.2.3 DeviceNet Power

This state is selected when the DeviceNet power is supplied through the DeviceNet connector on the side of the device. The COMM (PWR) selector switch must be set to **DNET**, and the Data Rate rotary switch must be set to **1, 2, 5,** or **P**.

10.3 DeviceNet Module Status LED (MOD LED)

The following table lists potential error conditions that can be detected by looking at MOD LED.

Module Status LED Color	Device State	Troubleshooting Comments
Flashing Red/Green persists indefinitely	Self Test Exception	If MFC device detected GFM Valve Leak Alarm while performing Self Test it will transition to the Self-Test Exception state. The device will remain in the Self-Test Exception state until operator intervention. Rerunning the GFM Valve Leak Check routine may clear this exception. Contact Field service support if problem does not go away after re-running the check
Flashing Green	Undefined	Semi-Sig DeviceNet state machines do not support this LED color Non-Semi-Sig compliant devices support this color, This condition may be an indication of broken LED
Flashing Red	Abort	Indicates that the normal operation of an MFC cannot be performed. Refer to diagnostic data. Request Alarm/Warning exception detail. NOTE: GFM Valve Leak Alarm if is detected during GFM Valve Check routine will put device into Abort State. The device will stay in the Abort state until operator intervention, or Service Reset request
	Baud Rate or MAC Id switch change detected	If Baud Rate or MAC ID switch change detected during operation, the device will continue to communicate using the MAC ID and baud rate assigned when the device was powered on. Once the device is power cycled, it will operate at the new switch settings If this condition was not triggered by the operator contact field service support.
Solid Red	Critical Fault	An Error state where operationally the device cannot recover In order to recover its normal operation, environmental conditions must be changed and the device is power cycled - Check Baud Rate switch /configuration to make sure it matches tool settings - Check DNET cable/connector for shorts or breaks

10.4 DeviceNet Network Status LED (NET LED)

The following table provides tips for troubleshooting GFC DeviceNet network problems.

Network Status LED Color	Network State	Indicated Behavior	Troubleshooting Comments
OFF	Not On-Line	Device has not completed its duplicate MAC ID check	- Normally indicates that GFC device is the only node on the network[1] . - If there are other devices on the network and Tool cannot establish communication with the MFC, check DNET Connector/Cable for breaks. Contact Field service if the issue cannot be resolved
	Link OK	One or more connections have been established	- If device is communicating with the tool, this condition may be an indication of broken LED.
Flashing Green	On-Line	All network connections are closed	- Check if Master is active on the network - Check if Master is configured to use the same baud rate as the GFC device - Check if Mac ID configured on the Device is in the master's scan list.
Solid Green	Link OK	One or more connections have been established[2]	- If the GFC device is not responding to commands, the master may have established communication on one connection but have tried to use another connection that has not been configured - Multiple connections could be writing to the same attributes[3]

[1] In order for the device to be active on the network, it must correctly process a duplicate MAC ID message. If there are no other nodes (master or slaves) on the network when the GFC device issues its duplicate MAC ID message, the device's network LED will remain off until such time as another node starts communicating on the network.

[2] Note: The GFC supports a single explicit connection and a single poll I/O connection. Having a solid green network LED indicates a poll I/O, an explicit, or both connections are active.

Network Status LED Color	Network State	Indicated Behavior	Troubleshooting Comments
Flashing Red	Connection Timed Out	Communication over one or more established connections has ceased to be processed	- GFC device is actively being managed in the master's scan list[4] - Master is communicating with the GFC device at the connection's configured EPR
Solid Red	Interface Failure	GFC device is no longer available on the network	- GFC devices' baud rate switch is set to the baud rate of the master - GFC devices' MAC ID is defined as unique value between 0 and 63 - No other device on the network has the same MAC ID as those assigned to GFC devices

[3] This is possible because, once established, poll I/O connections send data to a statically assigned group of attributes at a rate determined by the *expected packet rate* (EPR) of the connection. The user may try to use an explicit connection to simultaneously update these attributes being written by an active poll I/O connection. Under these conditions, attribute values from explicit connections will immediately be overwritten by values sent via the poll I/O.

[4] Connections can become inactive if the master does not continually update the connection.

10.5 NET and MOD Status LED Interaction (DNET mode only)

In general, the behavior indicated by the Network Status (NET) and Module Status (MOD) LEDs are defined independently. Under the following conditions, interactions between these two LEDs must be considered.

Module Status LED Color	Network Status LED Color	Indicated Behavior	Troubleshooting Comments
Solid Red	Solid Red	The GFC device detected an unrecoverable network error Under these conditions the MFC application has been transitioned to the Critical Fault state and all network communications have been disabled	- Check to see that GFC devices' baud rate switch is set to the baud rate of the master. - Ensure that GFC's MAC ID is defined as a unique value between 0 and 63. If Rate or Address switch is between detents, it will not read the correct value. Note that when switch is in detent, it may not line up perfectly with silkscreen on case.
Solid Green	Flashing Green	No connection created from Tool to this device	Check if MAC ID or Baud Rate are set correctly: - If rotary switch is between detents, it will not read the correct value. - Note that when switch is in detent, it may not line up perfectly with silkscreen on case.

10.6 RUN LED: GFC States

The following set of tables defines the GFC run states.

10.6.1 Calibration States

The following table defines the GFC warning state.

RUN LED Color	Condition	Troubleshooting Comments
Flashing Red	Normal: At the beginning of calibration process, during downstream pressure check	
	Error: After calibration is complete	GFC Leak is above alarm threshold. Try re-running either Calibration or Full Leak Check. If LED continues to flash red, replacement is suggested.
Solid Red	After calibration is complete	GFM Leak is above alarm threshold. Try re-running either Calibration or Full Leak Check. If LED continues to be solid red, GFC will not accept setpoints except 100% full scale for purge (in DNET mode, must be explicit command). GFC must be replaced.
Flashing Amber	Normal: During calibration and/or leak check	
Solid Amber	After calibration is complete	GFM or GFC Leak is above warning threshold. Try re-running either Calibration or Full Leak Check. If LED is still solid amber, monitor. Suggested replacement at next chamber PM.
Flashing Green	Normal: After calibration is done successfully	
	Calibration is canceled	If LED starts flashing green almost immediately after getting calibration request, this indicated that the calibration has been canceled. Check to see if tool sent Override command or a non-zero setpoint

10.6.2 Other Run LED States

RUN LED Color	Condition	Troubleshooting Comments
Flashing Green (cycles every 500ms)	Device is running fine	
Solid Green (not blinking for more than 1 minute)	Normal during GFC startup	
	During operation.	Software error. Power cycle to clear error Call customer support if error reoccurs
OFF	Normal during GFC startup	
	Application is not running	Software or hardware error if light does not appear after 5 minutes from Unit/Application restart. Power cycle to clear error (wait for 2–5 minutes after restart) Call customer support if error reoccurs
Solid Amber	During normal operation	One of the temperature sensors is reporting values outside thresholds. Check temperature conditions of GFC. Power-cycle device to attempt to clear error. Check Diagnostic Data tab of User Interface.
	During normal operation	GFM or GFC Valve Leak as measured by the Continuous Leak Check Routine is above warning threshold. Run Full Leak Check Routine
Blinking Amber	During GFC startup	Normal
	During calibration process	Normal
	During operation	May indicate reoccurring hardware problems Power-cycle device to clear error Call customer support

Blinking Red	During GFC startup	Normal during first 1 minute of operation
	During normal operation	GFM Valve Leak as measured by the Continuous Leak Check Routine is above Alarm threshold. Run Full Leak Check Routine.
Solid Red	After Leak Check Routine is complete	GFM Leak is above alarm threshold. Try re-running either Calibration or Full Leak Check. If LED continues to be solid red, GFC will not accept setpoints except 100% full scale for purge (in DNET mode, must be explicit command). GFC must be replaced.
	In most cases, unrecoverable error is detected (e.g., invalid internal configuration detected)	Critical Error Power cycle will not clear error Requires factory intervention. Send the unit back to manufacturer for error analysis